

Canonical W configurations - cluster P2

species: electron
polarization cluster: P2
source cluster configurations: 84
canonical configurations collected: 18

canonical display criterion: exactly 3 retained final-state components
retained-component cutoff: fraction $\geq 2.0\%$
normalized amplitude display: bar length = $|A_{\text{tilde}}|$, bar color/text = phase

cluster retained-component count distribution:

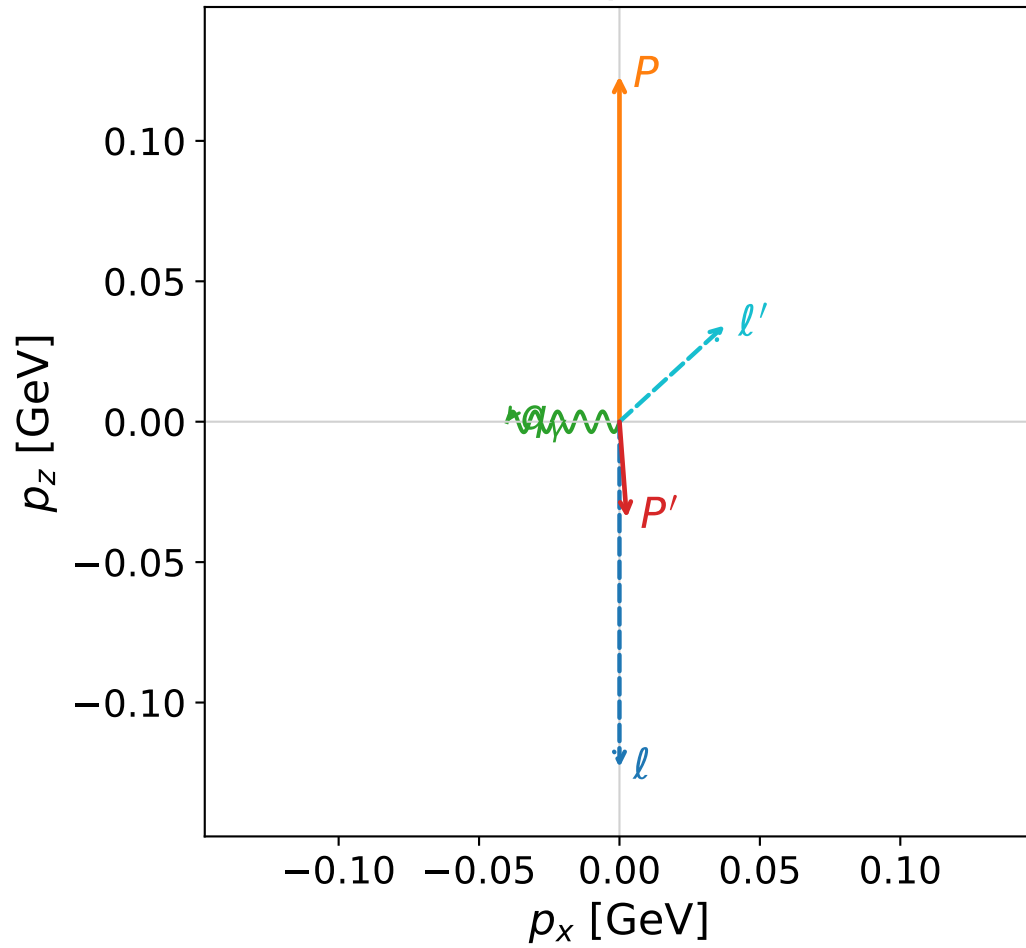
3 components:	18 configurations
4 components:	27 configurations
5 components:	7 configurations
6 components:	30 configurations
7 components:	2 configurations

CSV index:
Output\GradientPhaseSpaceScan\electron\Data\W\dw\polarization_cluster_02\combined\canonical_dw_polarization_cluster_02_configurations_electron.csv

Each following page is one complete momentum, kinematic-summary, and normalized-amplitude configuration.

coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

z--x plane



dw_mixing_angles_polarization_02_local_minimum_379

$D_W=0.00121395$

region: polarization_cluster_02_local_minimum_379

outgoing-state purity: 1

kinematic point: gradient_local_minimum_0379

lepton mixing angle: $\alpha_e=0.79117094$ rad

incoming lepton: $0.703013|+\rangle + 0.711177|-\rangle$

proton mixing angle: $\alpha_p=1.7484036$ rad

incoming proton: $-0.176675|+\rangle + 0.984269|-\rangle$

$s=1.14298$, $\sqrt{s}=1.0691$, $|\vec{P}|=0.123061$, $|\vec{P}'|=0.0348211$

$\theta_{p'}=2.95231$, $\theta_\gamma=1.5721$, $\phi_{p'}=5.11662$, $\phi_\gamma=2.25205$

$E_\gamma=0.063631$, $\theta_{l'}=1.03207$, $\phi_{l'}=5.42489$

$Q^2=0.0248839$, $x_B=0.171454$, $t=-0.0247191$, $W^2=1.00009$, $y=0.551566$

four-momenta $[E, p_x, p_y, p_z]$ GeV:

l $E= 0.1231$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1231$

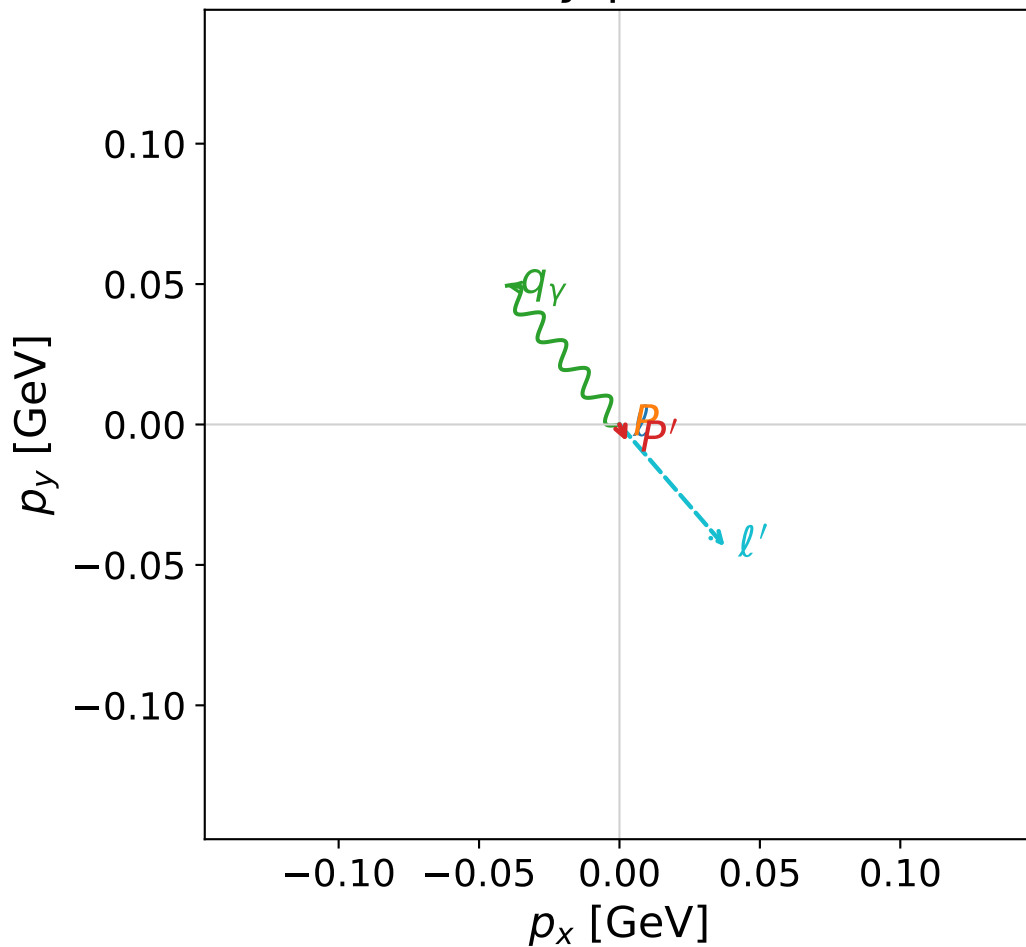
P $E= 0.9460$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1231$

l' $E= 0.0668$ $p_x= 0.0375$ $p_y= -0.0434$ $p_z= 0.0343$

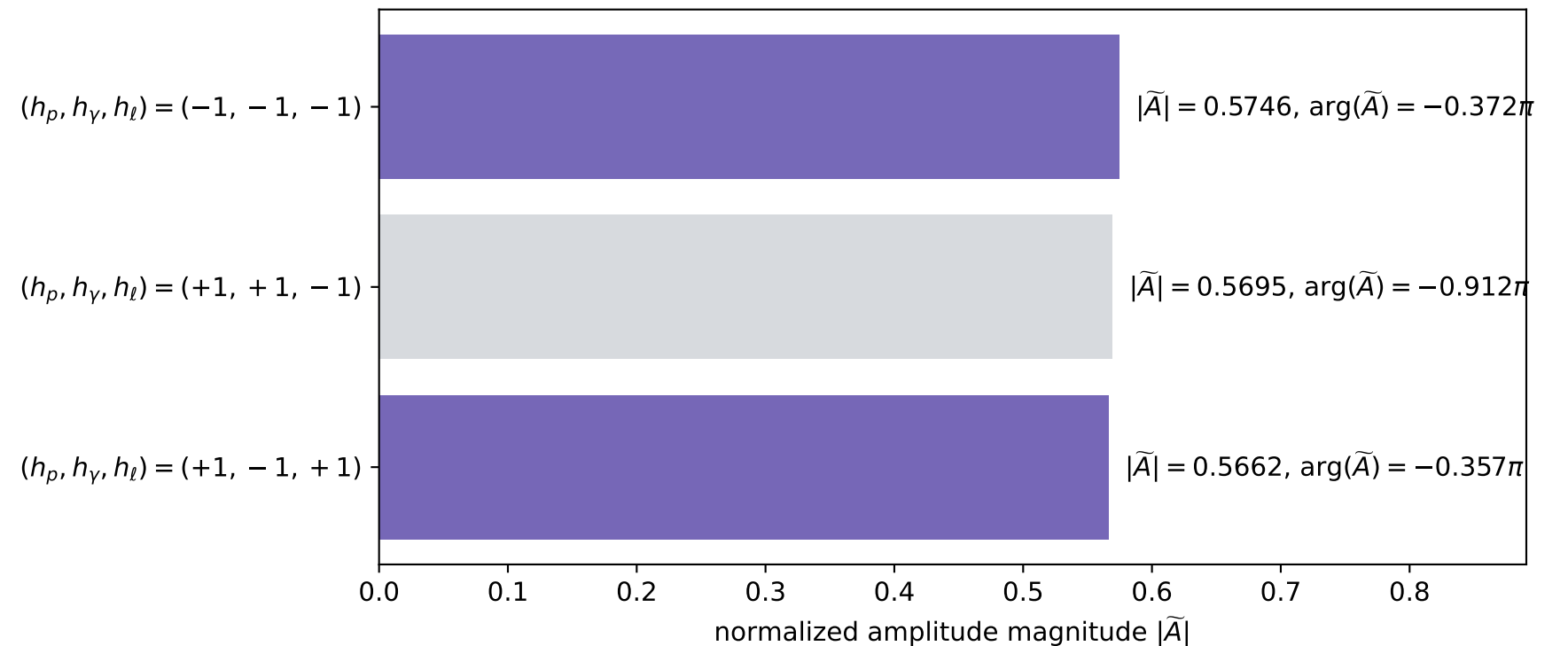
P' $E= 0.9386$ $p_x= 0.0026$ $p_y= -0.0060$ $p_z= -0.0342$

q_γ $E= 0.0636$ $p_x= -0.0401$ $p_y= 0.0494$ $p_z= -0.0001$

x--y plane

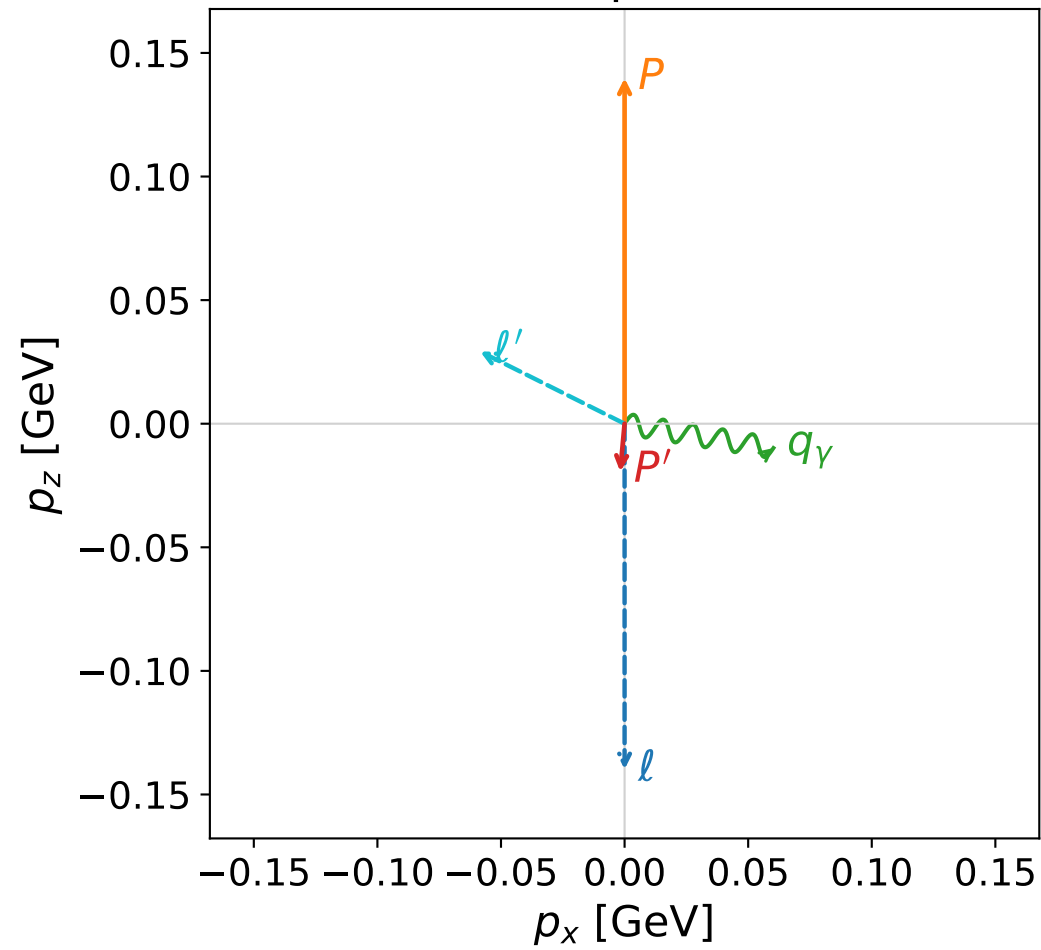


Leading normalized final-state amplitudes
(N=3, retained=97.5%)

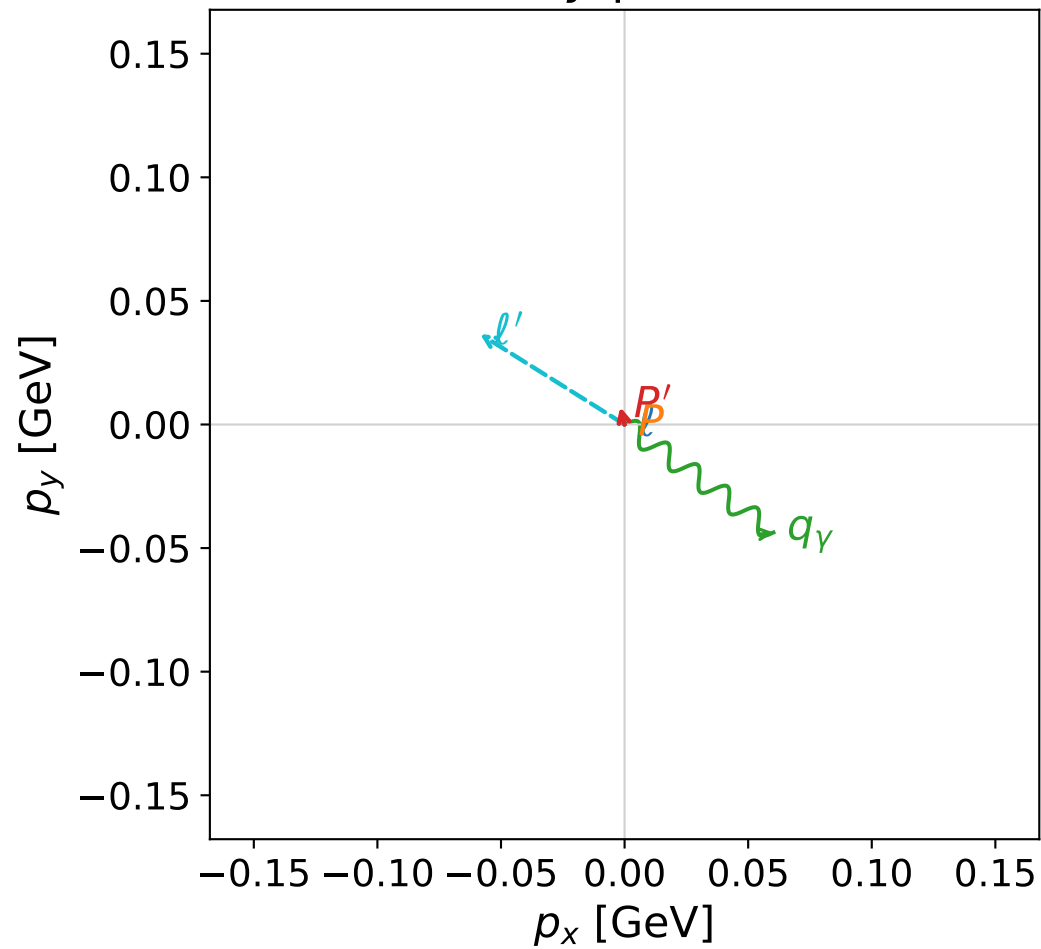


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

z--x plane



x--y plane

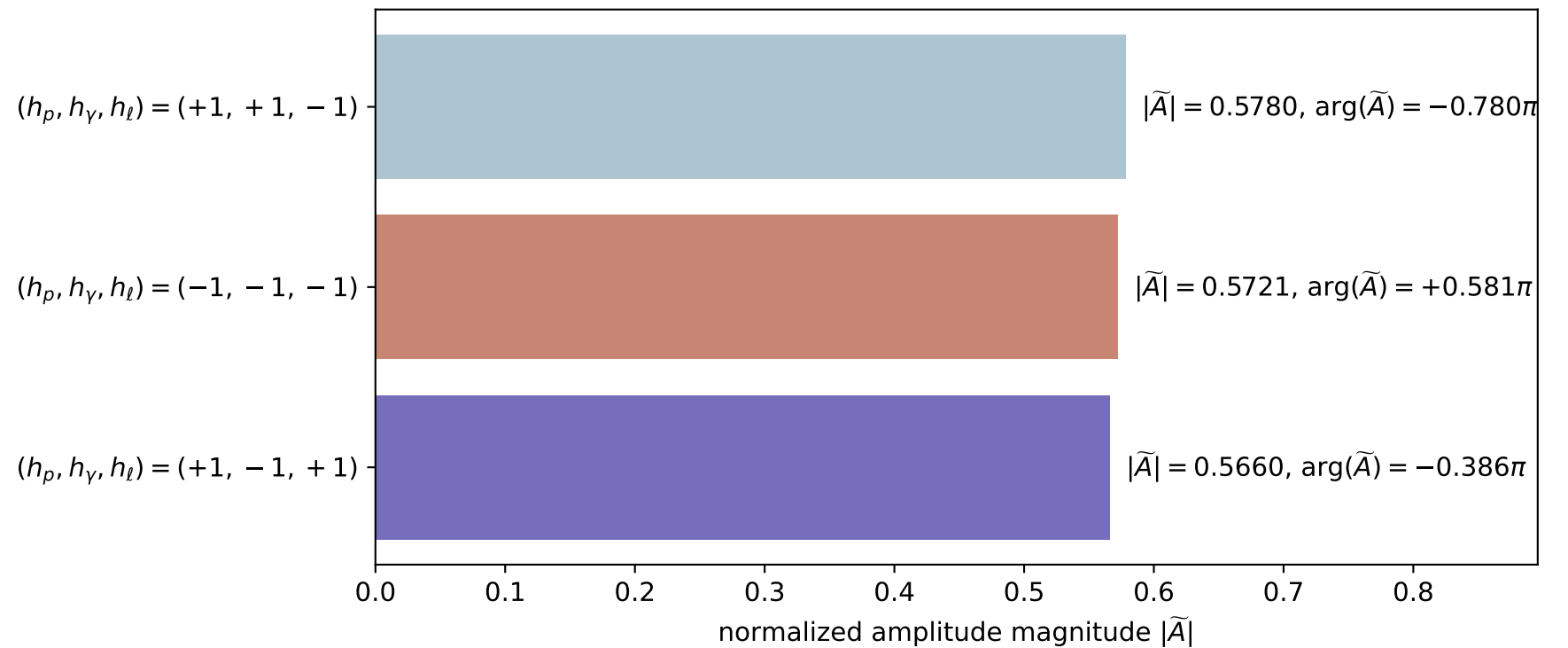


dw_mixing_angles_polarization_02_local_minimum_442
 $D_W=0.00176181$
 region: polarization_cluster_02_local_minimum_442
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0442
 lepton mixing angle: $\alpha_e=0.80580383$ rad
 incoming lepton: $0.692532|+\rangle + 0.721388|-\rangle$
 proton mixing angle: $\alpha_p=1.3763217$ rad
 incoming proton: $0.193251|+\rangle + 0.981149|-\rangle$

$s=1.18413$, $\sqrt{s}=1.08818$, $|\vec{P}|=0.139813$, $|\vec{P}'|=0.0206987$
 $\theta_{p'}=2.78097$, $\theta_\gamma=1.70231$, $\phi_{p'}=1.81532$, $\phi_\gamma=5.65562$
 $E_\gamma=0.0750484$, $\theta_{l'}=1.17019$, $\phi_{l'}=2.5822$
 $Q^2=0.0291113$, $x_B=0.170854$, $t=-0.0252891$, $W^2=1.02112$, $y=0.55996$

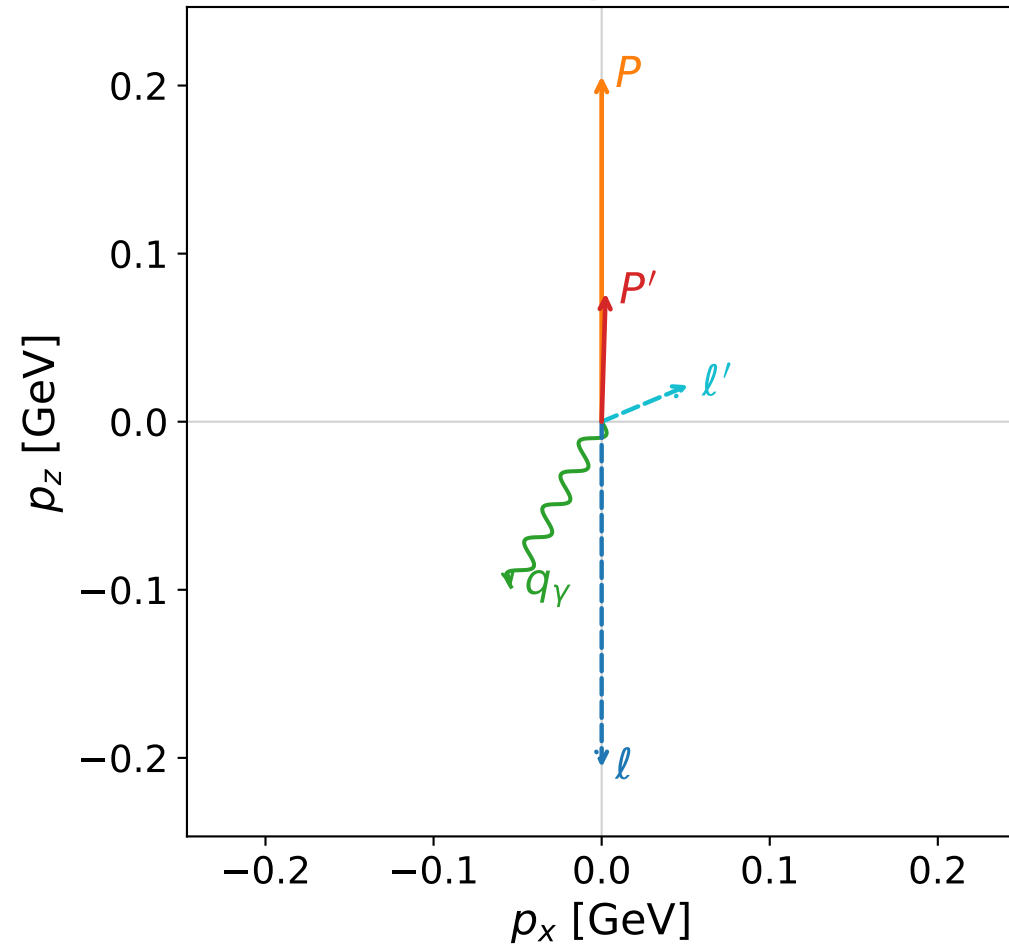
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 0.1398$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1398$
 P $E= 0.9484$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1398$
 l' $E= 0.0749$ $p_x= -0.0585$ $p_y= 0.0366$ $p_z= 0.0292$
 P' $E= 0.9382$ $p_x= -0.0018$ $p_y= 0.0071$ $p_z= -0.0194$
 q_γ $E= 0.0750$ $p_x= 0.0602$ $p_y= -0.0437$ $p_z= -0.0098$

Leading normalized final-state amplitudes
(N=3, retained=98.2%)



coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

z--x plane

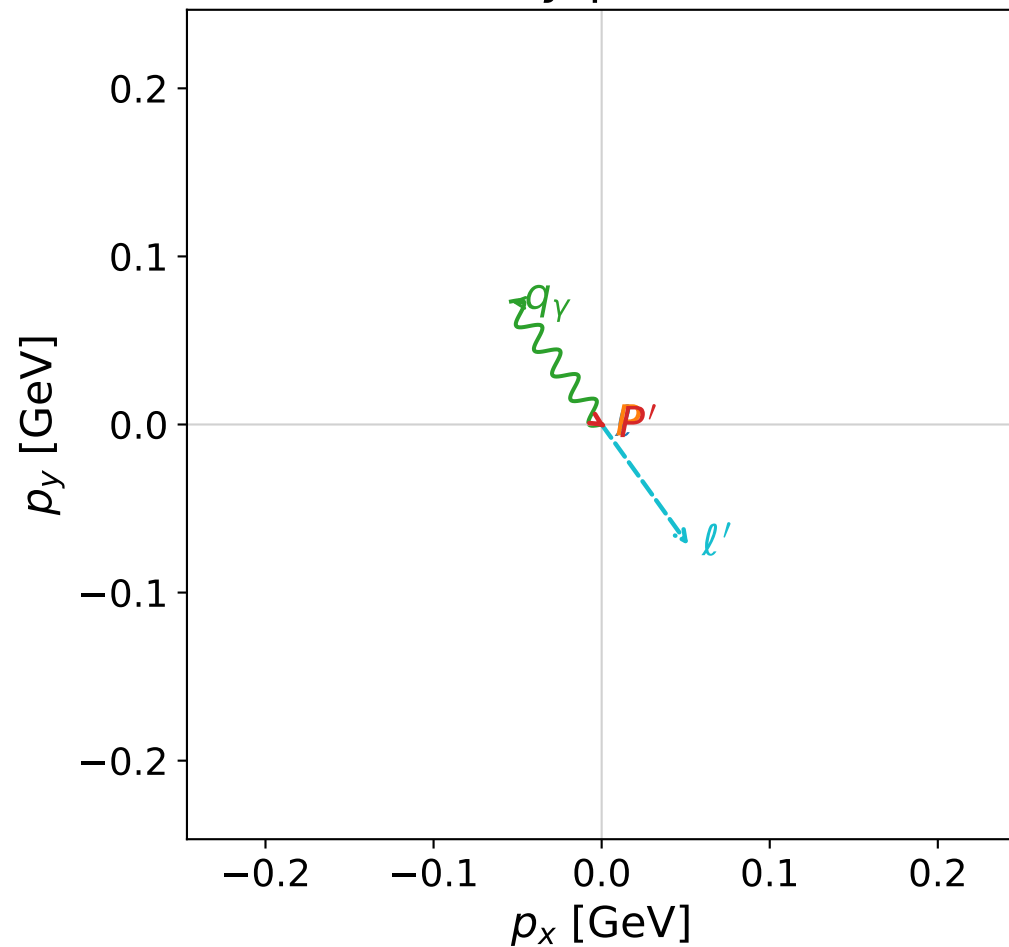


dw_mixing_angles_polarization_02_local_minimum_590
 $D_W=0.00392354$
 region: polarization_cluster_02_local_minimum_590
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0590
 lepton mixing angle: $\alpha_e=0.78087704$ rad
 incoming lepton: $0.710296|+\rangle + 0.703903|-\rangle$
 proton mixing angle: $\alpha_p=1.6435266$ rad
 incoming proton: $-0.0726662|+\rangle + 0.997356|-\rangle$

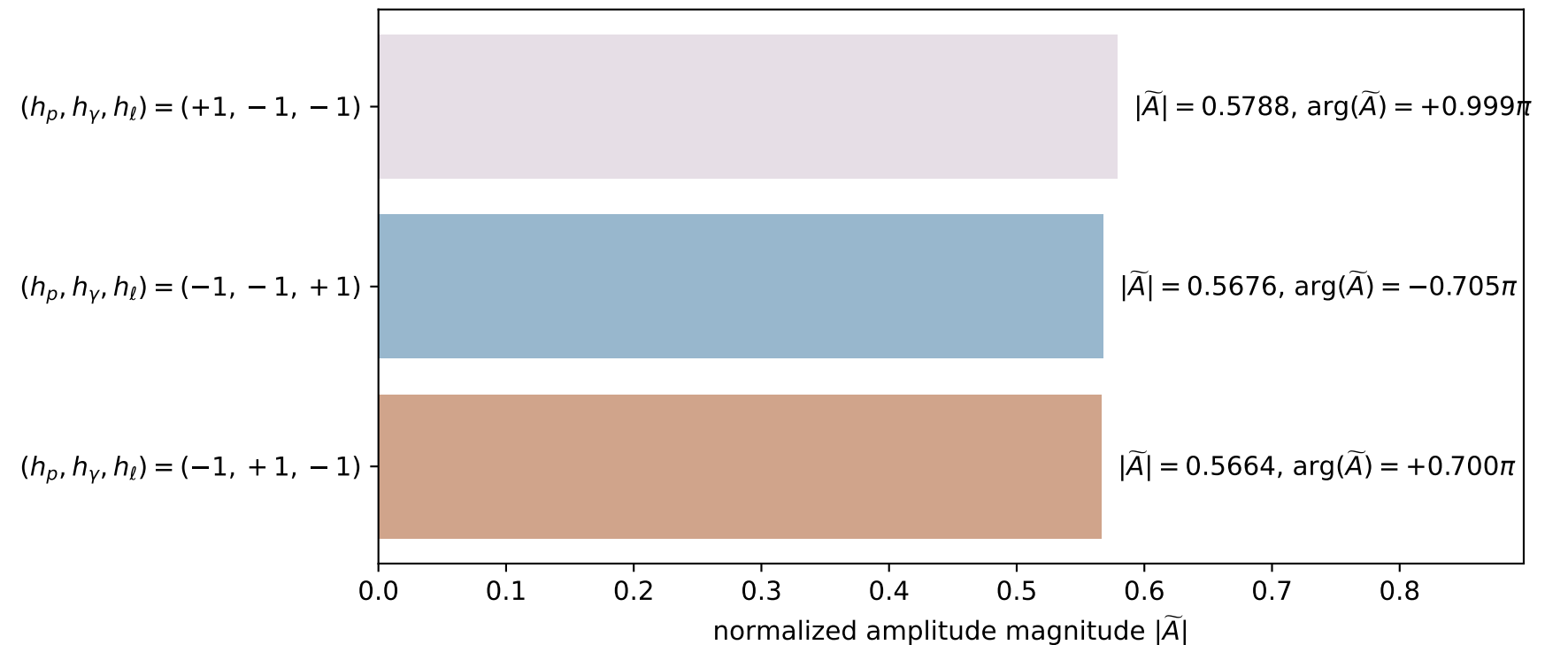
$s=1.35928$, $\sqrt{s}=1.16588$, $|\vec{P}|=0.205609$, $|\vec{P}'|=0.0765314$
 $\theta_{p'}=0.036252$, $\theta_\gamma=2.39442$, $\phi_{p'}=5.72703$, $\phi_\gamma=2.20606$
 $E_\gamma=0.133793$, $\theta_{l'}=1.33026$, $\phi_{l'}=5.33602$
 $Q^2=0.0463199$, $x_B=0.147689$, $t=-0.0163149$, $W^2=1.14715$, $y=0.654171$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 0.2056$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.2056$
 P $E= 0.9603$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.2056$
 l' $E= 0.0910$ $p_x= 0.0516$ $p_y= -0.0717$ $p_z= 0.0217$
 P' $E= 0.9411$ $p_x= 0.0024$ $p_y= -0.0015$ $p_z= 0.0765$
 q_γ $E= 0.1338$ $p_x= -0.0540$ $p_y= 0.0732$ $p_z= -0.0982$

x--y plane

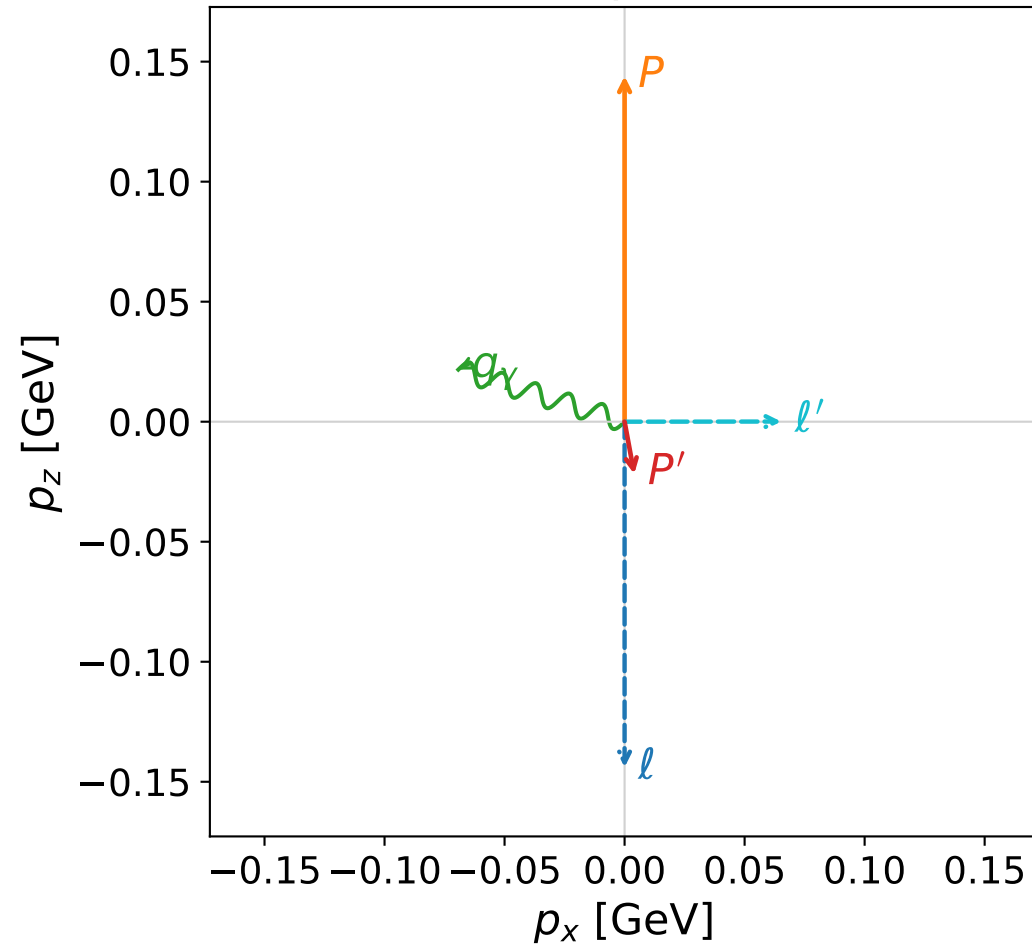


Leading normalized final-state amplitudes
 (N=3, retained=97.8%)



coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

z--x plane

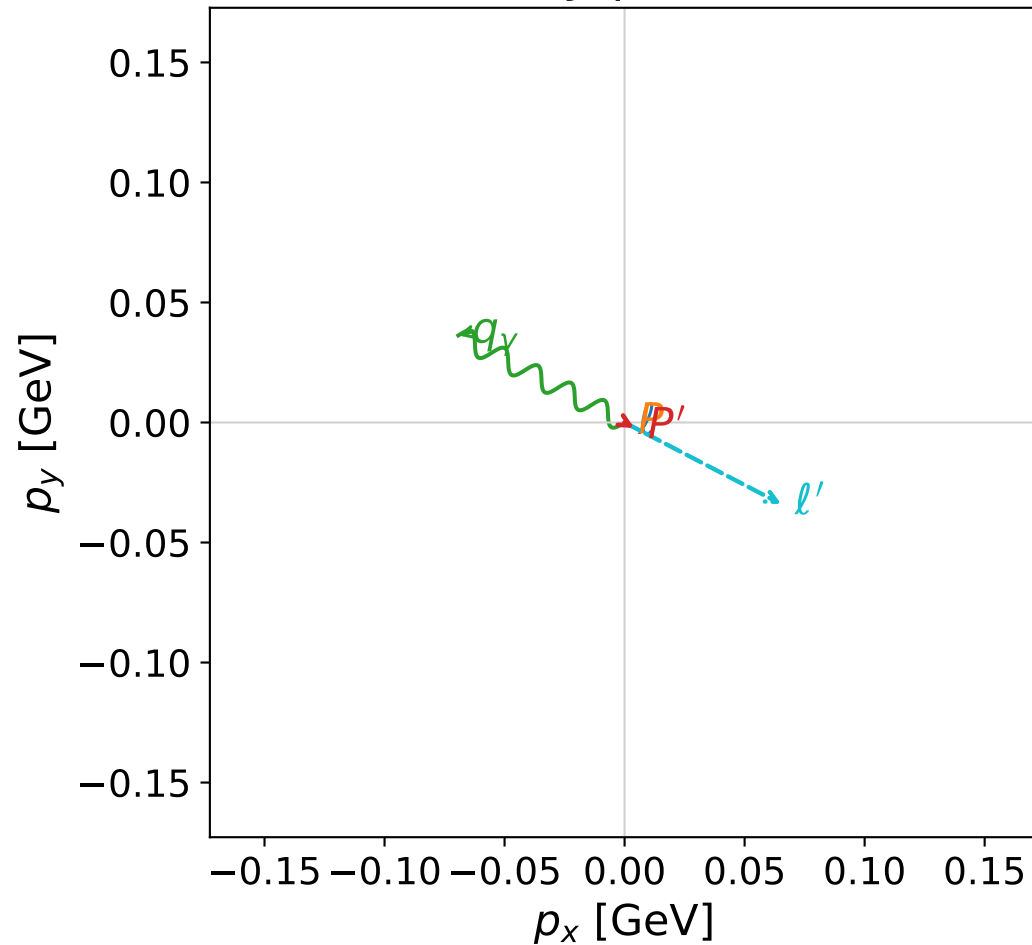


dw_mixing_angles_polarization_02_local_minimum_592
 $D_W=0.00392984$
 region: polarization_cluster_02_local_minimum_592
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0592
 lepton mixing angle: $\alpha_e=0.77959771$ rad
 incoming lepton: $0.711196|+\rangle +0.702993|-\rangle$
 proton mixing angle: $\alpha_p=1.6772067$ rad
 incoming proton: $-0.10621|+\rangle +0.994344|-\rangle$

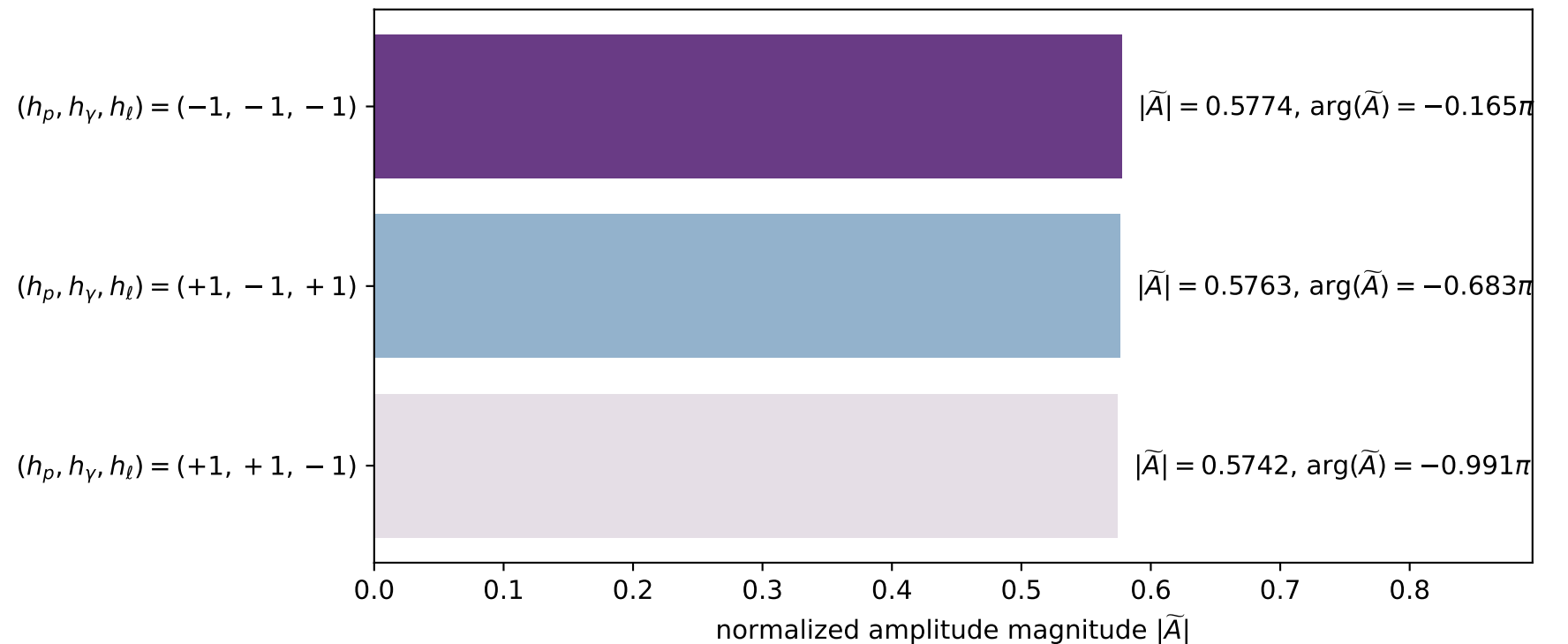
$s=1.19455$, $\sqrt{s}=1.09296$, $|\vec{P}|=0.14397$, $|\vec{P}'|=0.022253$
 $\theta_{p'}=2.93211$, $\theta_\gamma=1.30008$, $\phi_{p'}=5.76646$, $\phi_\gamma=2.65937$
 $E_\gamma=0.0811348$, $\theta_{l'}=1.56985$, $\phi_{l'}=5.80313$
 $Q^2=0.0211997$, $x_B=0.121059$, $t=-0.0273752$, $W^2=1.03376$, $y=0.55645$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 0.1440$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1440$
 P $E= 0.9490$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1440$
 l' $E= 0.0736$ $p_x= 0.0652$ $p_y= -0.0340$ $p_z= 0.0001$
 P' $E= 0.9383$ $p_x= 0.0040$ $p_y= -0.0023$ $p_z= -0.0218$
 q_γ $E= 0.0811$ $p_x= -0.0693$ $p_y= 0.0363$ $p_z= 0.0217$

x--y plane

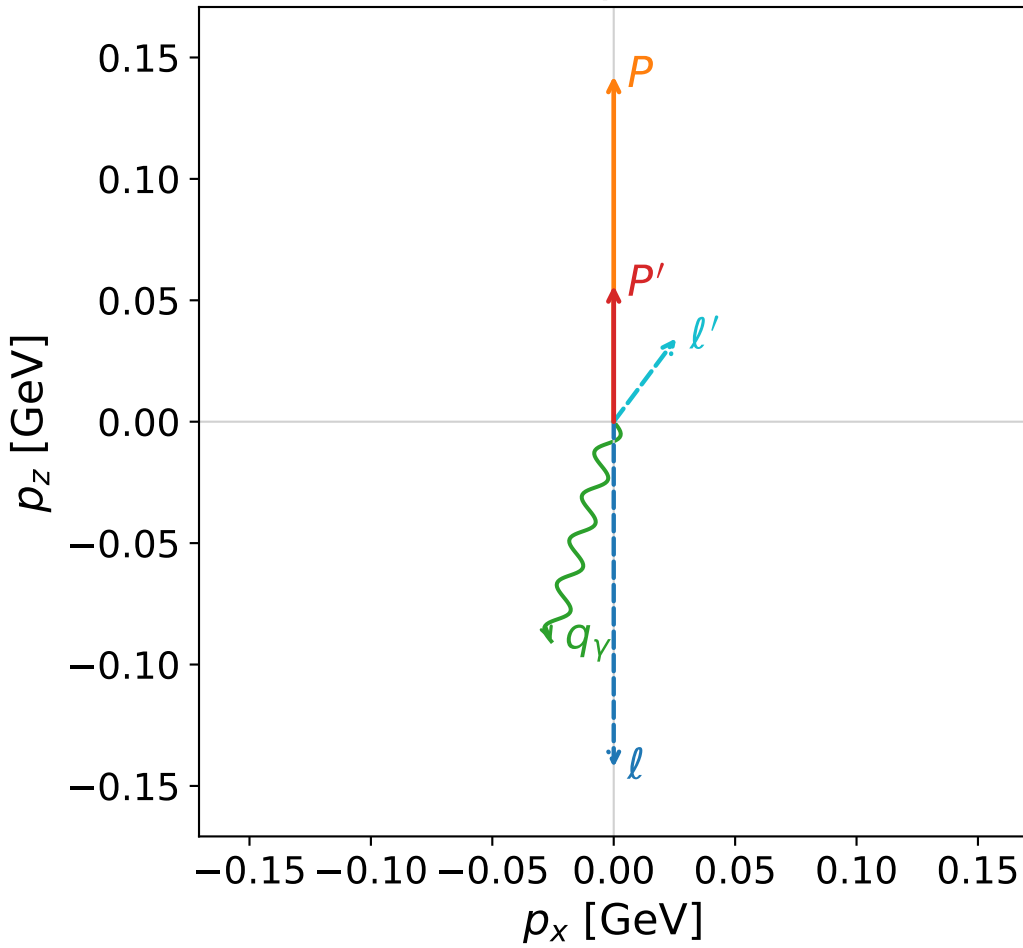


Leading normalized final-state amplitudes
(N=3, retained=99.5%)

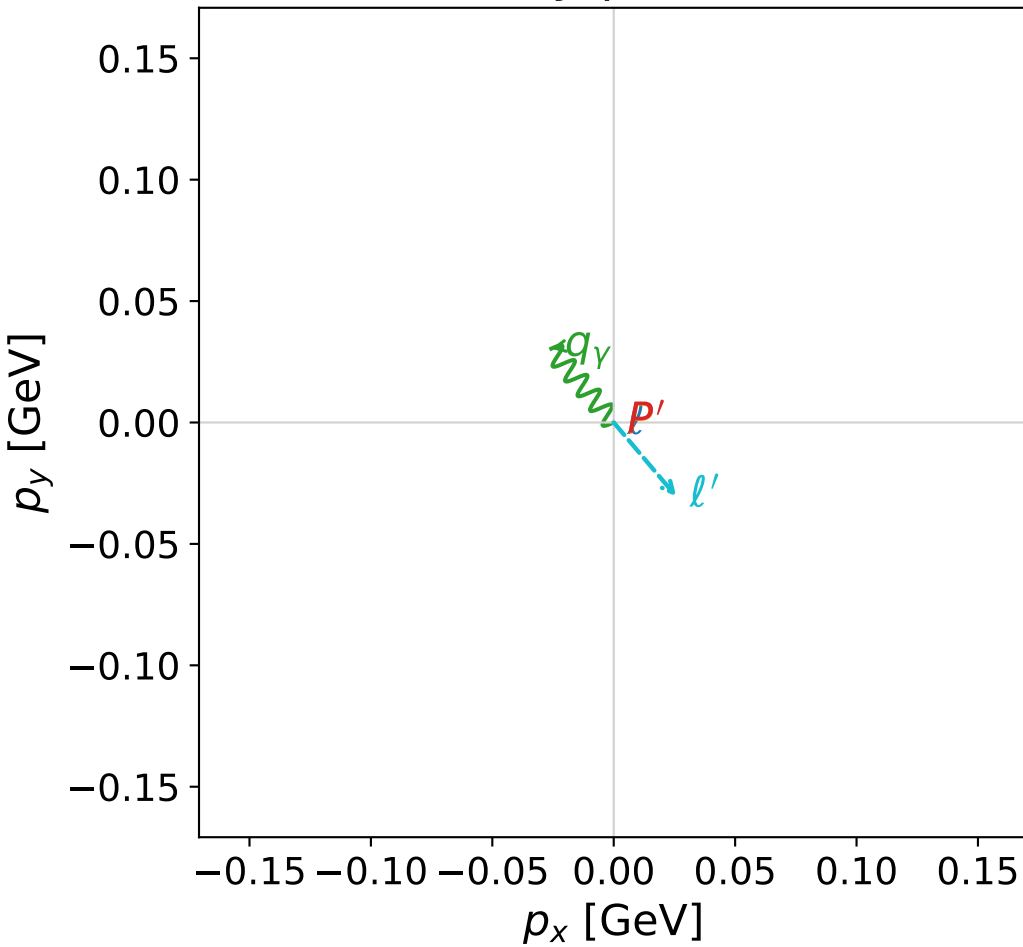


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

z--x plane



x--y plane



dw_mixing_angles_polarization_02_local_minimum_693

$D_W=0.00687147$

region: polarization_cluster_02_local_minimum_693

outgoing-state purity: 1

kinematic point: gradient_local_minimum_0693

lepton mixing angle: $\alpha_e=0.78199713$ rad

incoming lepton: $0.709508|+\rangle +0.704698|-\rangle$

proton mixing angle: $\alpha_p=1.6520513$ rad

incoming proton: $-0.0811656|+\rangle +0.996701|-\rangle$

$s=1.19042$, $\sqrt{s}=1.09106$, $|\vec{P}|=0.142324$, $|\vec{P}'|=0.0560568$

$\theta_{P'}=0$, $\theta_\gamma=2.72655$, $\phi_{P'}=0.420211$, $\phi_\gamma=2.2728$

$E_\gamma=0.0987891$, $\theta_{l'}=0.859265$, $\phi_{l'}=5.41439$

$Q^2=0.0247481$, $x_B=0.112214$, $t=-0.00735997$, $W^2=1.07564$, $y=0.710119$

four-momenta $[E, p_x, p_y, p_z]$ GeV:

l $E= 0.1423$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1423$

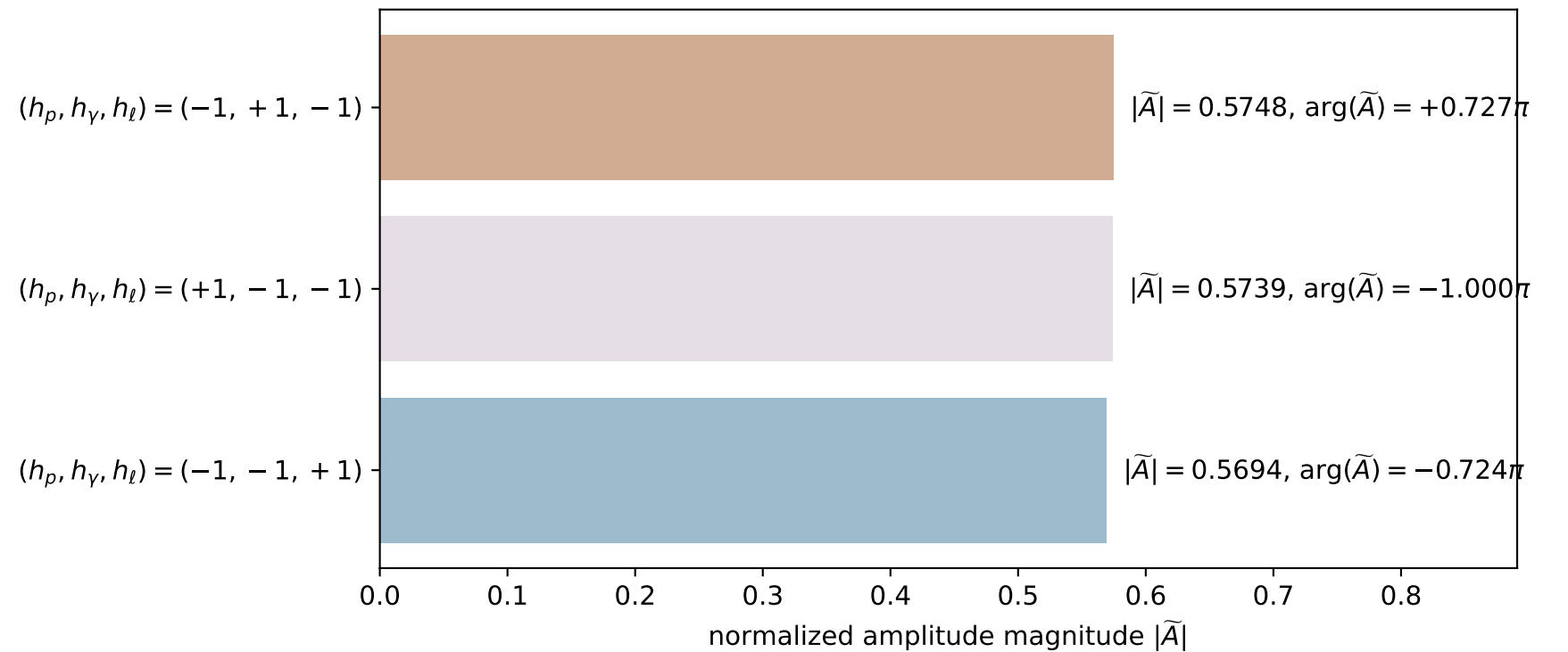
P $E= 0.9487$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1423$

l' $E= 0.0526$ $p_x= 0.0257$ $p_y= -0.0304$ $p_z= 0.0343$

P' $E= 0.9397$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.0561$

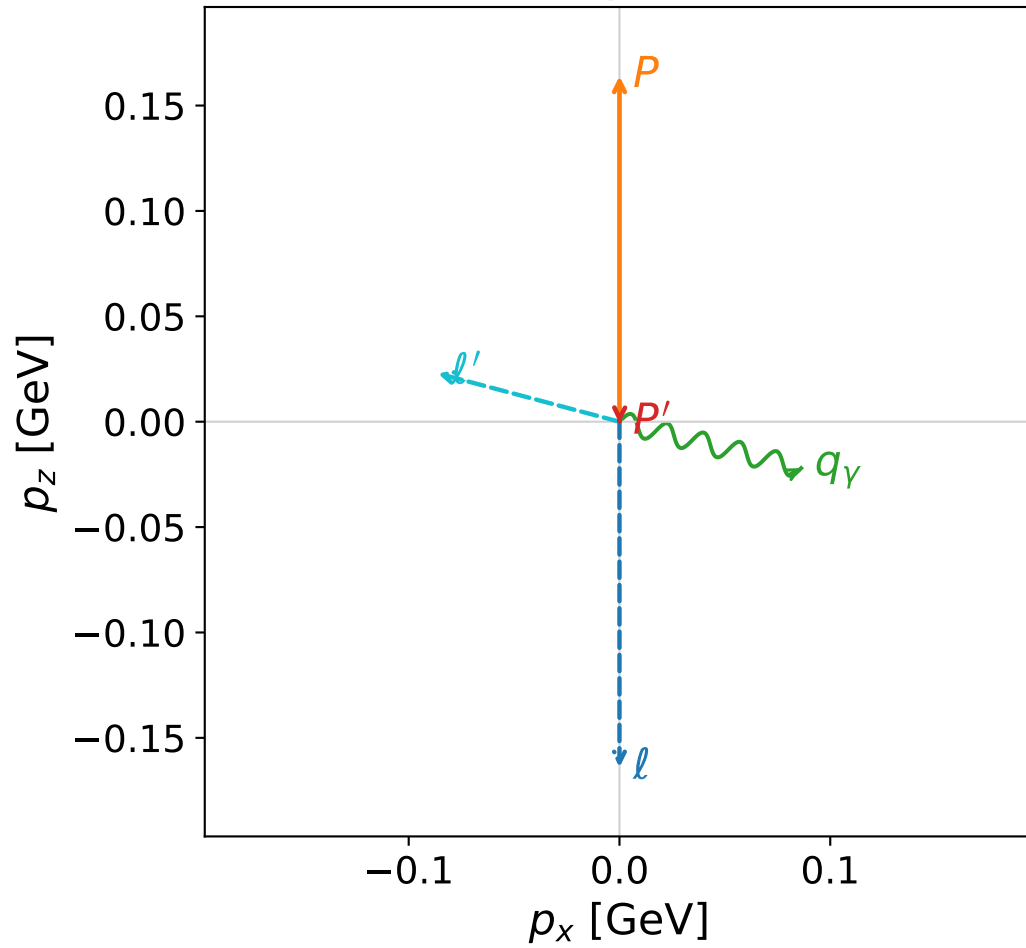
q_γ $E= 0.0988$ $p_x= -0.0257$ $p_y= 0.0304$ $p_z= -0.0904$

Leading normalized final-state amplitudes
(N=3, retained=98.4%)



coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

z--x plane

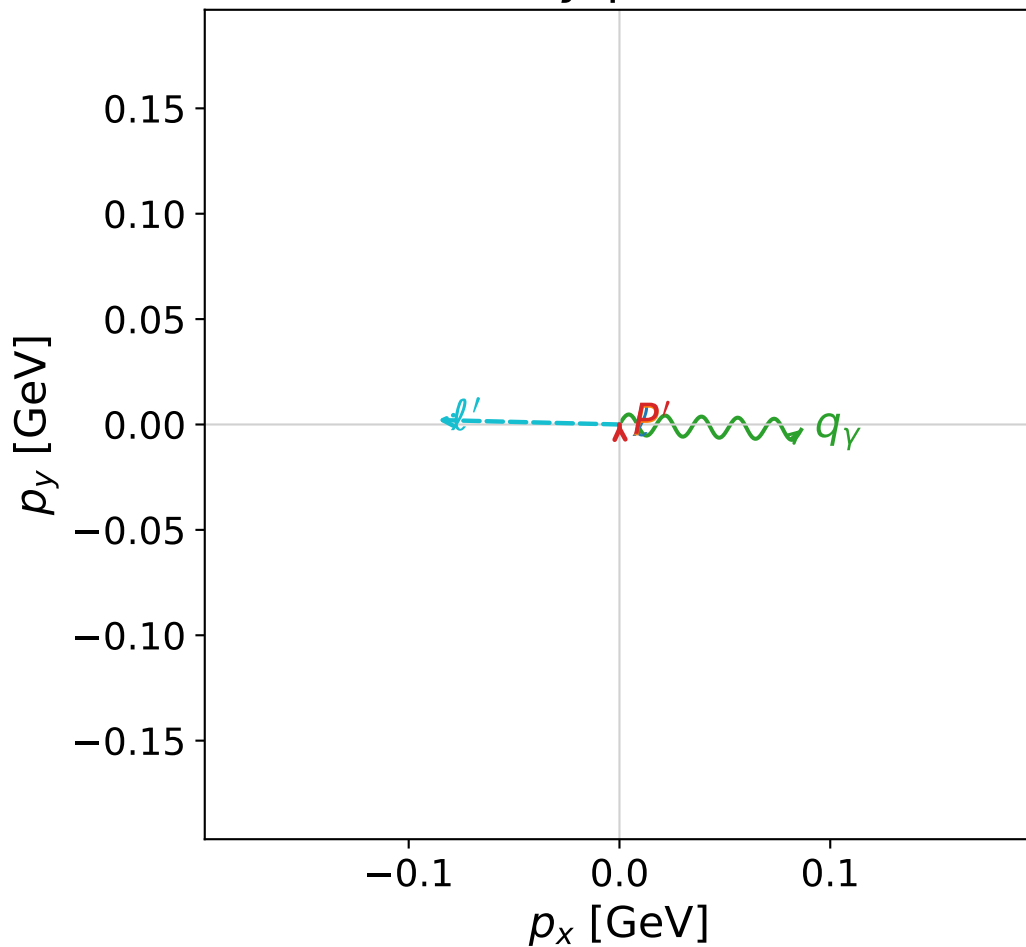


dw_mixing_angles_polarization_02_local_minimum_713
 $D_W=0.00788043$
 region: polarization_cluster_02_local_minimum_713
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0713
 lepton mixing angle: $\alpha_e=0.80739396$ rad
 incoming lepton: $0.691384|+\rangle + 0.722488|-\rangle$
 proton mixing angle: $\alpha_p=1.3762109$ rad
 incoming proton: $0.19336|+\rangle + 0.981128|-\rangle$

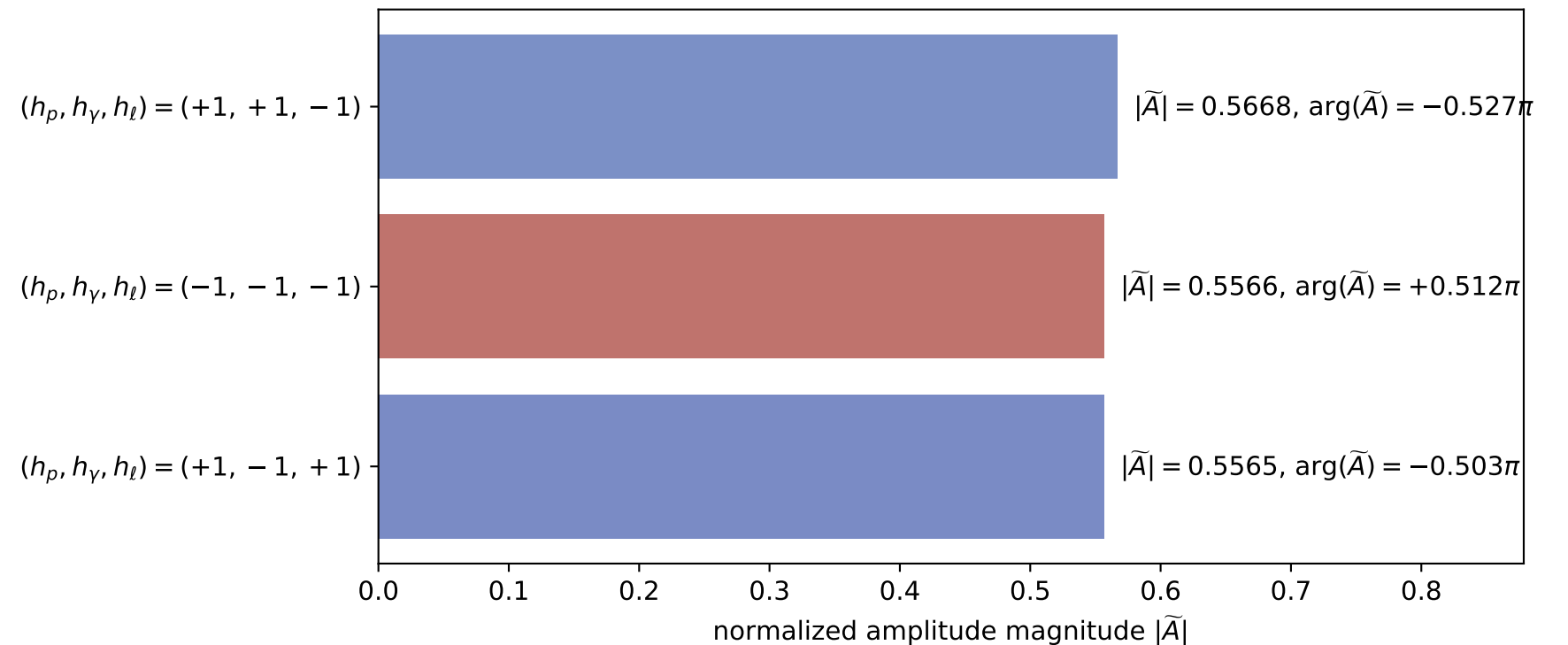
$s=1.24588$, $\sqrt{s}=1.11619$, $|\vec{P}|=0.163967$, $|\vec{P}'|=0.000947889$
 $\theta_{p'}=2.69809$, $\theta_\gamma=1.82098$, $\phi_{p'}=1.60646$, $\phi_\gamma=6.25378$
 $E_\gamma=0.0889989$, $\theta_{\ell'}=1.31124$, $\phi_{\ell'}=3.11689$
 $Q^2=0.0367552$, $x_B=0.180452$, $t=-0.0269645$, $W^2=1.04677$, $y=0.556456$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1640$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1640$
 P $E= 0.9522$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1640$
 ℓ' $E= 0.0892$ $p_x= -0.0862$ $p_y= 0.0021$ $p_z= 0.0229$
 P' $E= 0.9380$ $p_x= -0.0000$ $p_y= 0.0004$ $p_z= -0.0009$
 q_γ $E= 0.0890$ $p_x= 0.0862$ $p_y= -0.0025$ $p_z= -0.0220$

x--y plane

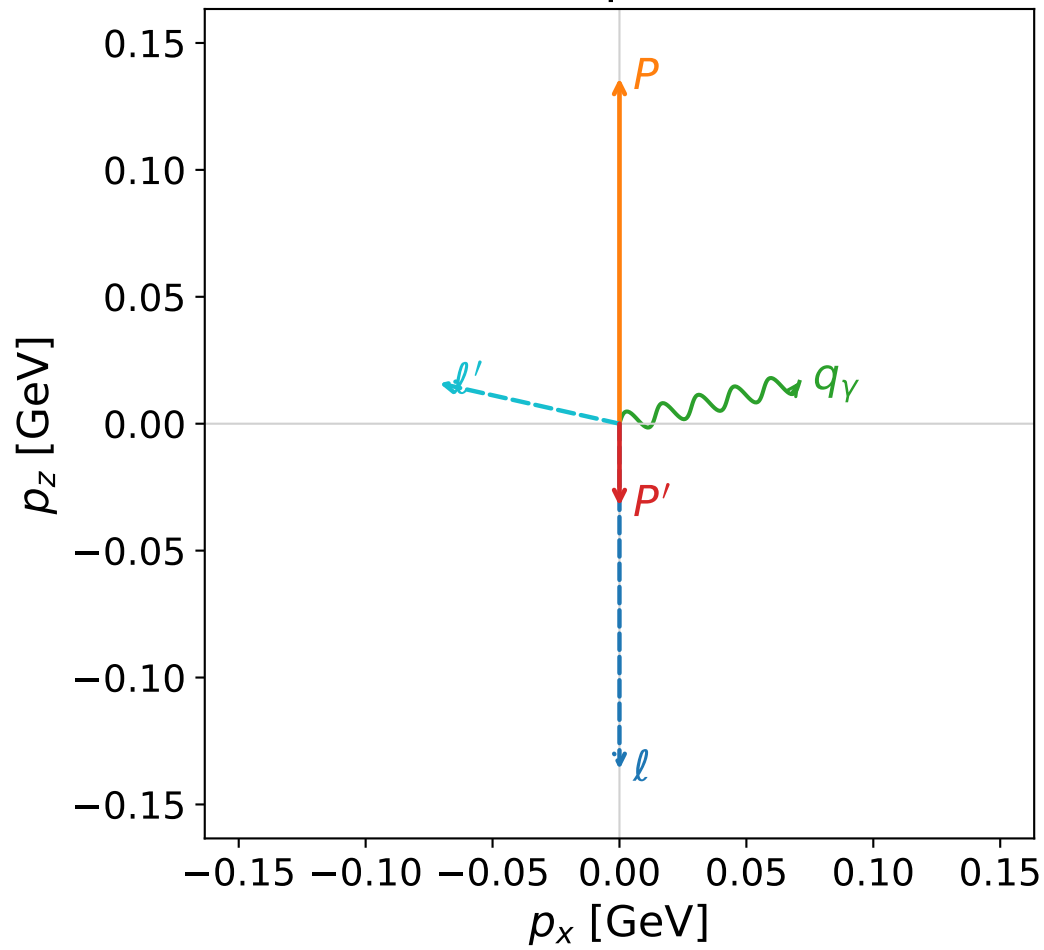


Leading normalized final-state amplitudes
 (N=3, retained=94.1%)

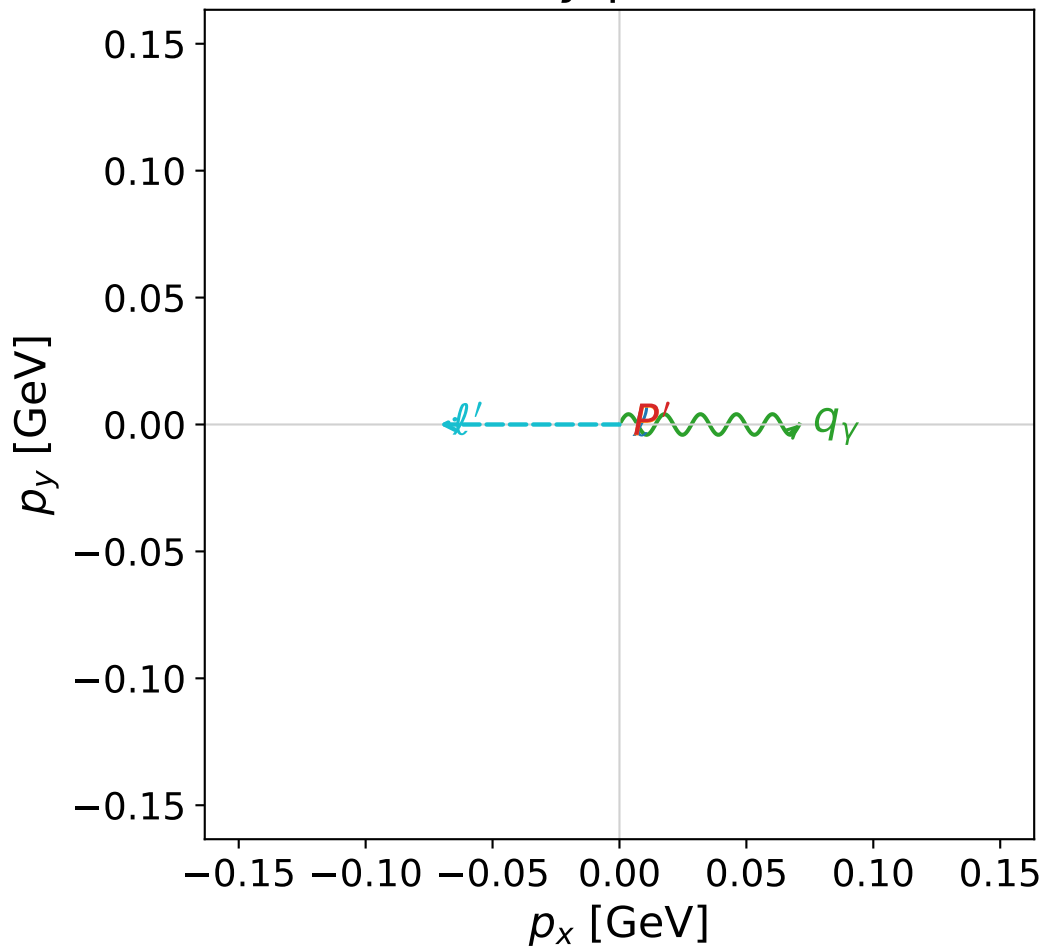


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

z--x plane



x--y plane

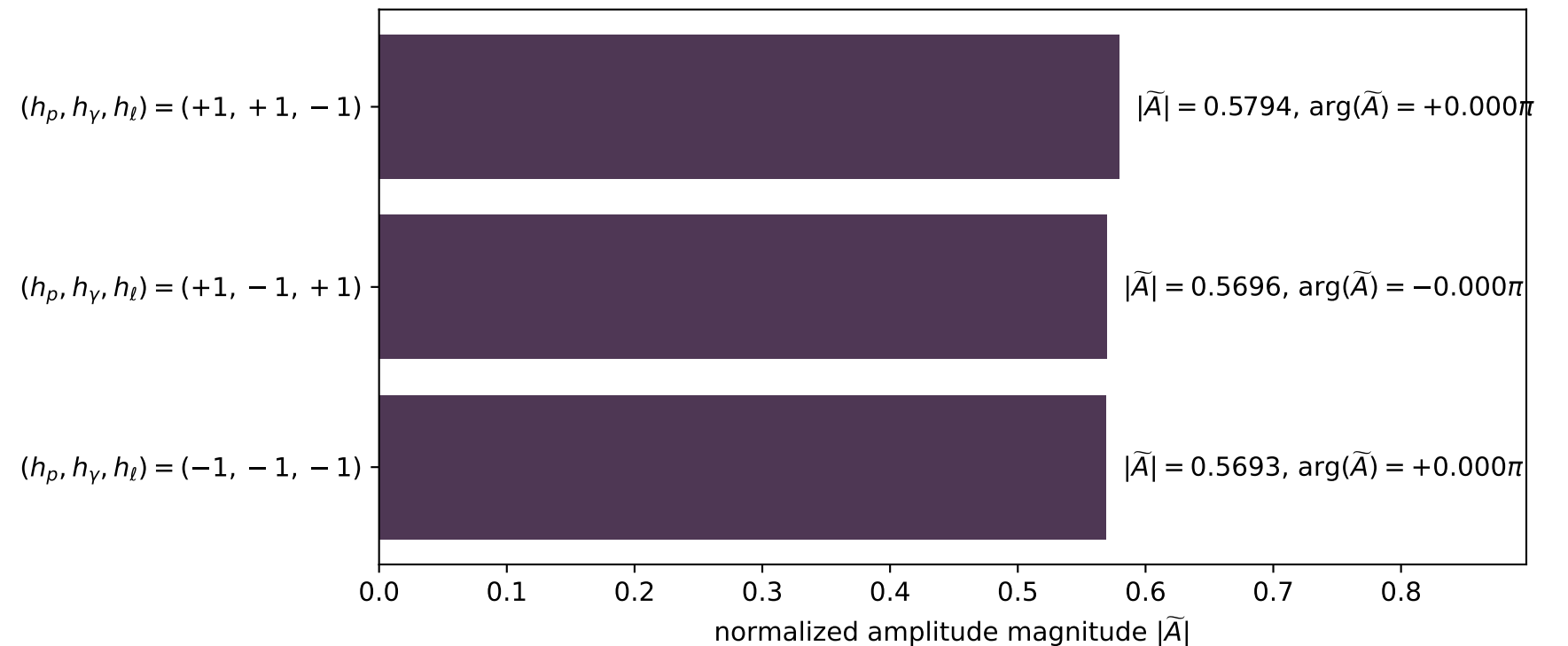


dw_mixing_angles_polarization_02_local_minimum_726
 $D_W=0.00815666$
 region: polarization_cluster_02_local_minimum_726
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0726
 lepton mixing angle: $\alpha_e=0.80690904$ rad
 incoming lepton: $0.691734|+\rangle + 0.722153|-\rangle$
 proton mixing angle: $\alpha_p=1.3699192$ rad
 incoming proton: $0.199529|+\rangle + 0.979892|-\rangle$

$s=1.17502$, $\sqrt{s}=1.08398$, $|\vec{P}|=0.136152$, $|\vec{P}'|=0.0323792$
 $\theta_{p'}=3.14159$, $\theta_\gamma=1.34233$, $\phi_{p'}=4.50934$, $\phi_\gamma=7.32617e-06$
 $E_\gamma=0.0727764$, $\theta_{\ell'}=1.35019$, $\phi_{\ell'}=3.1416$
 $Q^2=0.0241106$, $x_B=0.149026$, $t=-0.0283168$, $W^2=1.01752$, $y=0.548109$

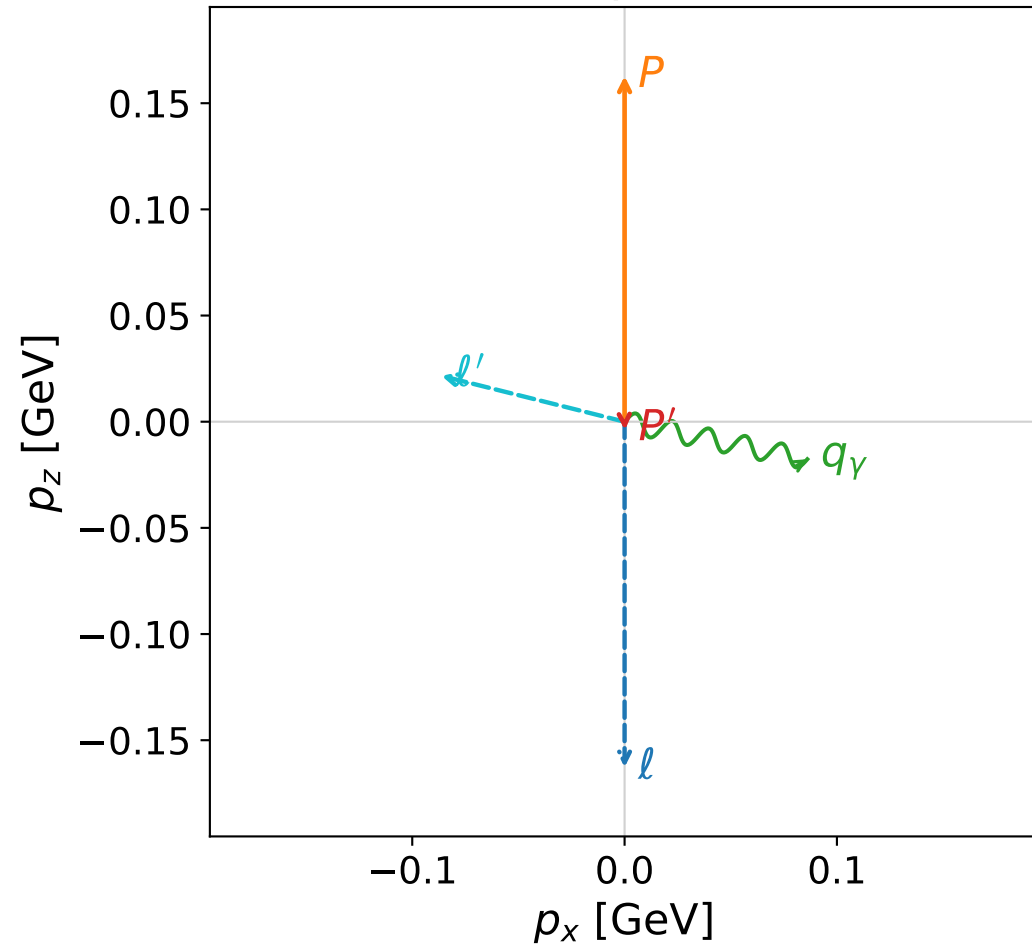
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1362$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1362$
 P $E= 0.9478$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1362$
 ℓ' $E= 0.0726$ $p_x= -0.0709$ $p_y= -0.0000$ $p_z= 0.0159$
 P' $E= 0.9386$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -0.0324$
 q_γ $E= 0.0728$ $p_x= 0.0709$ $p_y= 0.0000$ $p_z= 0.0165$

Leading normalized final-state amplitudes
 (N=3, retained=98.4%)



coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

z--x plane

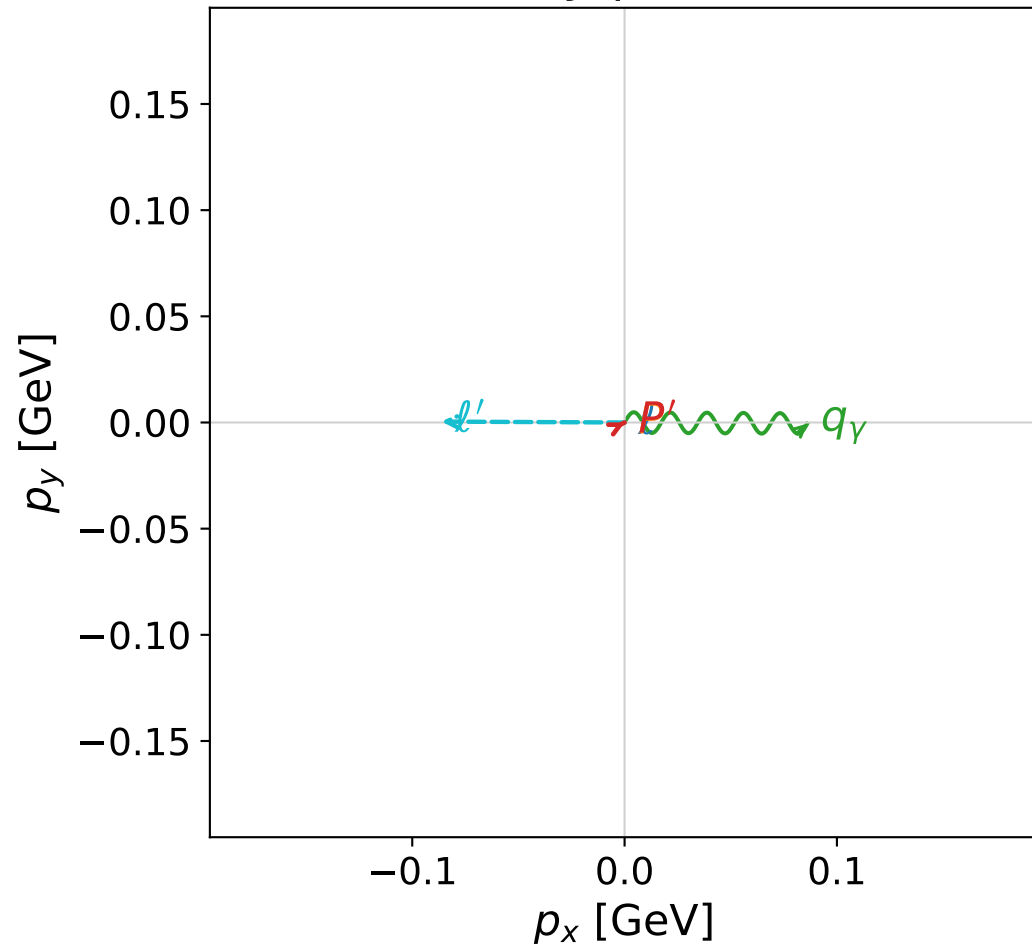


dw_mixing_angles_polarization_02_local_minimum_729
 $D_W=0.00833546$
 region: polarization_cluster_02_local_minimum_729
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0729
 lepton mixing angle: $\alpha_e=0.80706385$ rad
 incoming lepton: $0.691622|+\rangle + 0.72226|-\rangle$
 proton mixing angle: $\alpha_p=1.3715292$ rad
 incoming proton: $0.197951|+\rangle + 0.980212|-\rangle$

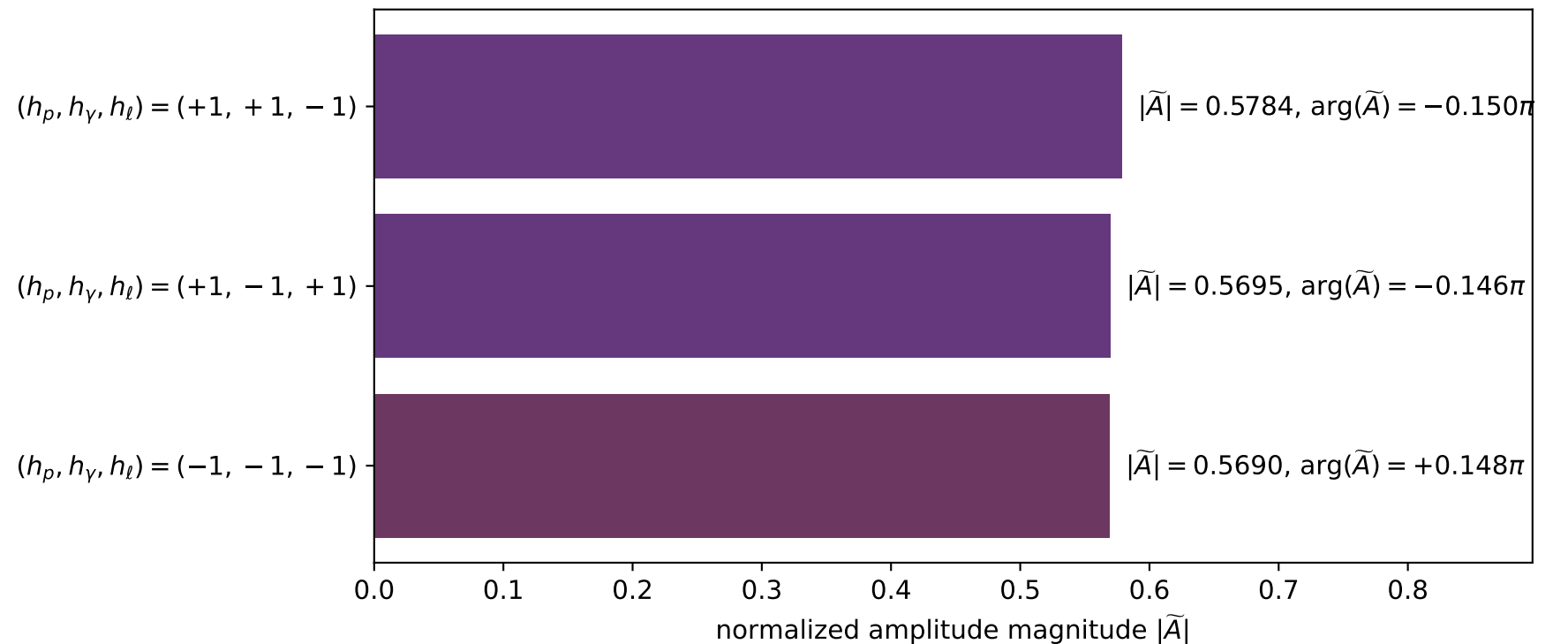
$s=1.24272$, $\sqrt{s}=1.11477$, $|\vec{P}|=0.162758$, $|\vec{P}'|=0.00392398$
 $\theta_{p'}=3.09273$, $\theta_\gamma=1.77317$, $\phi_{p'}=0.464333$, $\phi_\gamma=6.27731$
 $E_\gamma=0.087864$, $\theta_{l'}=1.32561$, $\phi_{l'}=3.13672$
 $Q^2=0.0359632$, $x_B=0.179252$, $t=-0.027585$, $W^2=1.04451$, $y=0.552885$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E=0.1628$ $p_x=0.0000$ $p_y=0.0000$ $p_z=-0.1628$
 P $E=0.9520$ $p_x=0.0000$ $p_y=0.0000$ $p_z=0.1628$
 l' $E=0.0889$ $p_x=-0.0862$ $p_y=0.0004$ $p_z=0.0216$
 P' $E=0.9380$ $p_x=0.0002$ $p_y=0.0001$ $p_z=-0.0039$
 q_γ $E=0.0879$ $p_x=0.0861$ $p_y=-0.0005$ $p_z=-0.0177$

x--y plane

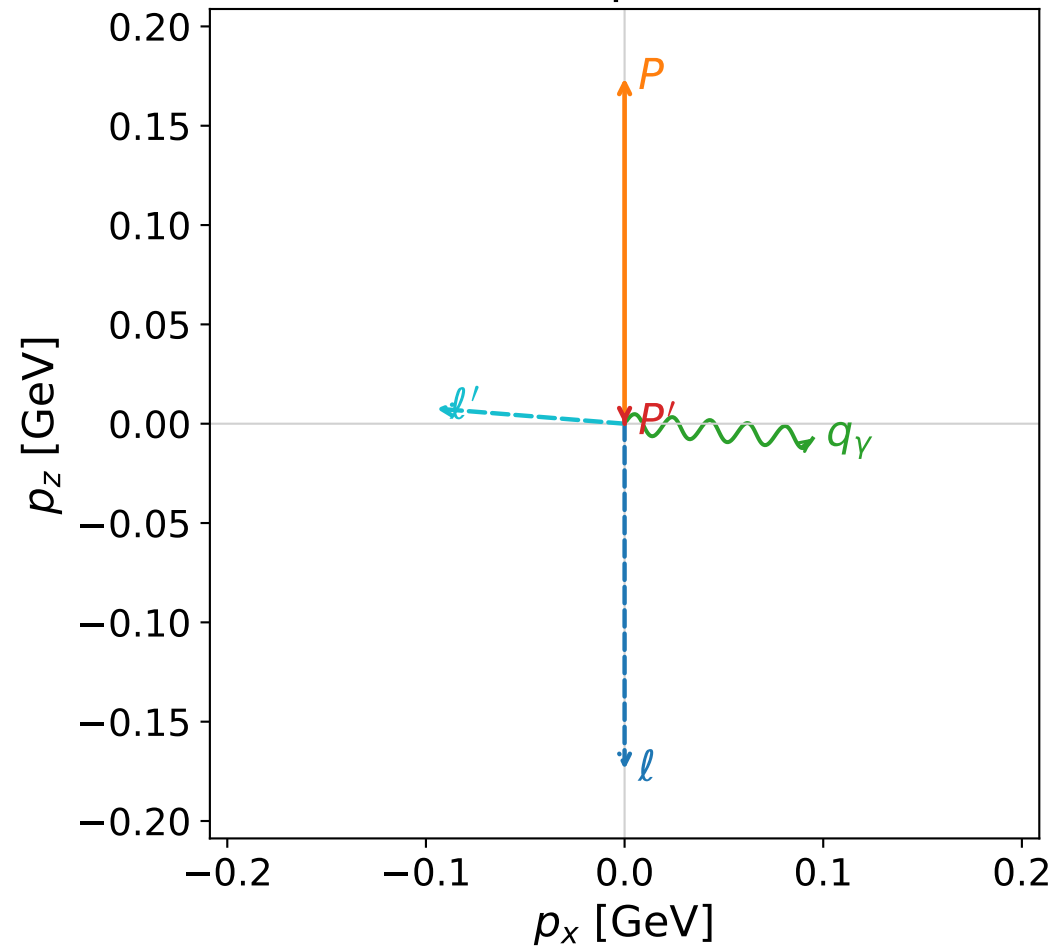


Leading normalized final-state amplitudes
(N=3, retained=98.3%)



coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

z--x plane

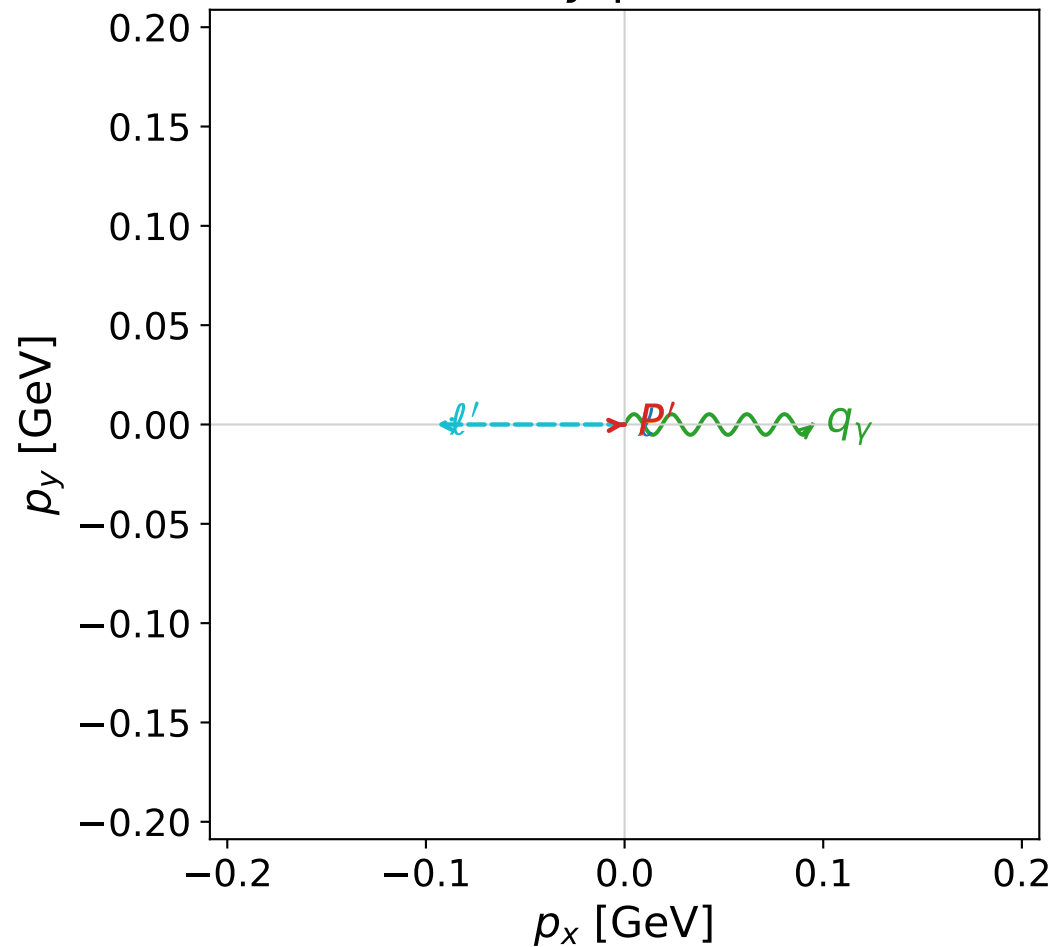


dw_mixing_angles_polarization_02_local_minimum_739
 $D_W=0.0085628$
 region: polarization_cluster_02_local_minimum_739
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0739
 lepton mixing angle: $\alpha_e=0.7888655$ rad
 incoming lepton: $0.704651|+\rangle +0.709554|-\rangle$
 proton mixing angle: $\alpha_p=1.3637025$ rad
 incoming proton: $0.205617|+\rangle +0.978633|-\rangle$

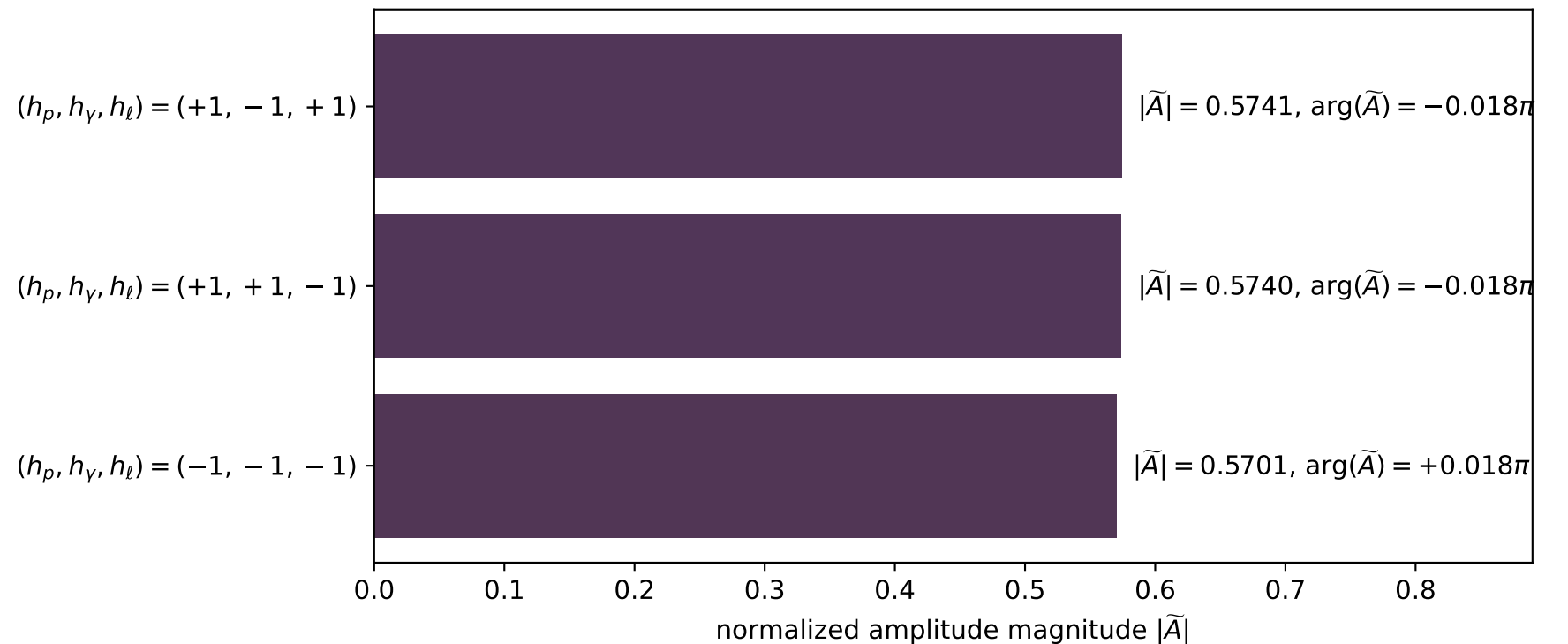
$s=1.27231$, $\sqrt{s}=1.12797$, $|\vec{P}|=0.173971$, $|\vec{P}'|=0.000134712$
 $\theta_{p'}=3.14126$, $\theta_\gamma=1.64947$, $\phi_{p'}=0.0552426$, $\phi_\gamma=1.07418e-06$
 $E_\gamma=0.0949782$, $\theta_{l'}=1.49071$, $\phi_{l'}=3.14159$
 $Q^2=0.0356947$, $x_B=0.166898$, $t=-0.0300569$, $W^2=1.05802$, $y=0.54494$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 0.1740$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1740$
 P $E= 0.9540$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1740$
 l' $E= 0.0950$ $p_x= -0.0947$ $p_y= -0.0000$ $p_z= 0.0076$
 P' $E= 0.9380$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.0001$
 q_γ $E= 0.0950$ $p_x= 0.0947$ $p_y= 0.0000$ $p_z= -0.0075$

x--y plane

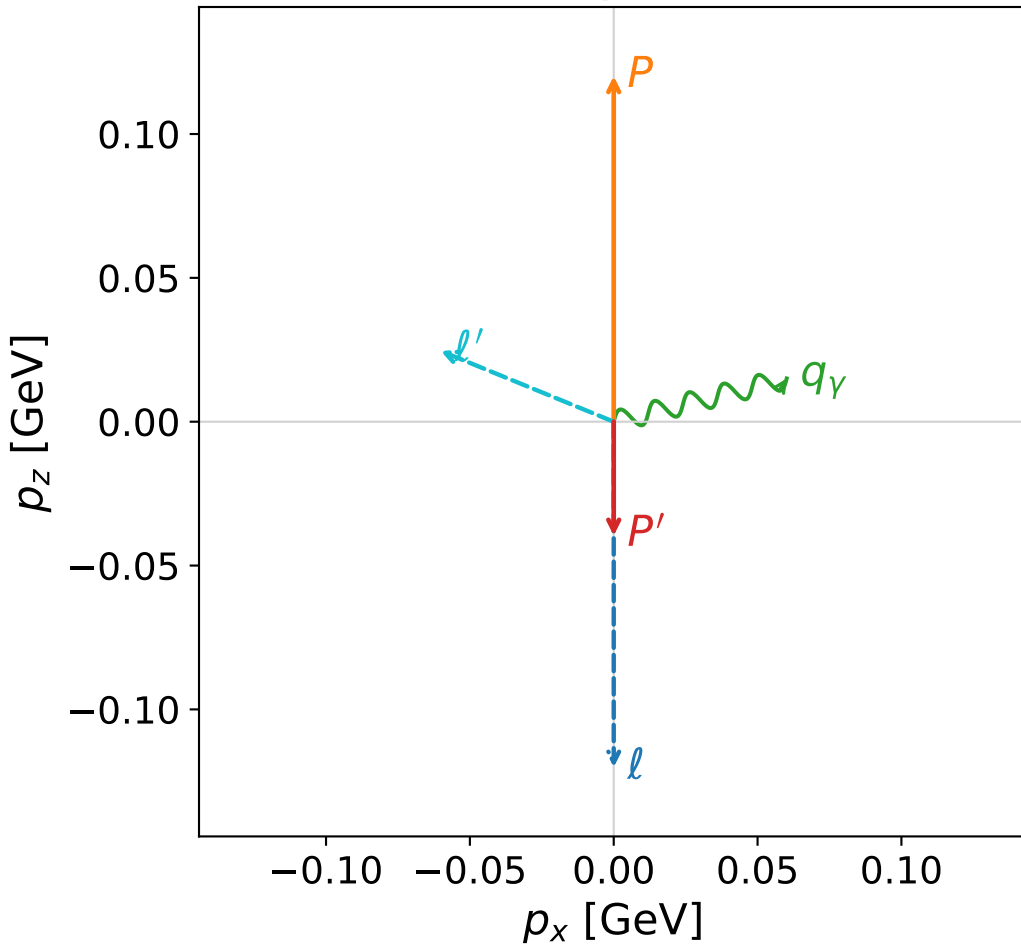


Leading normalized final-state amplitudes
 (N=3, retained=98.4%)

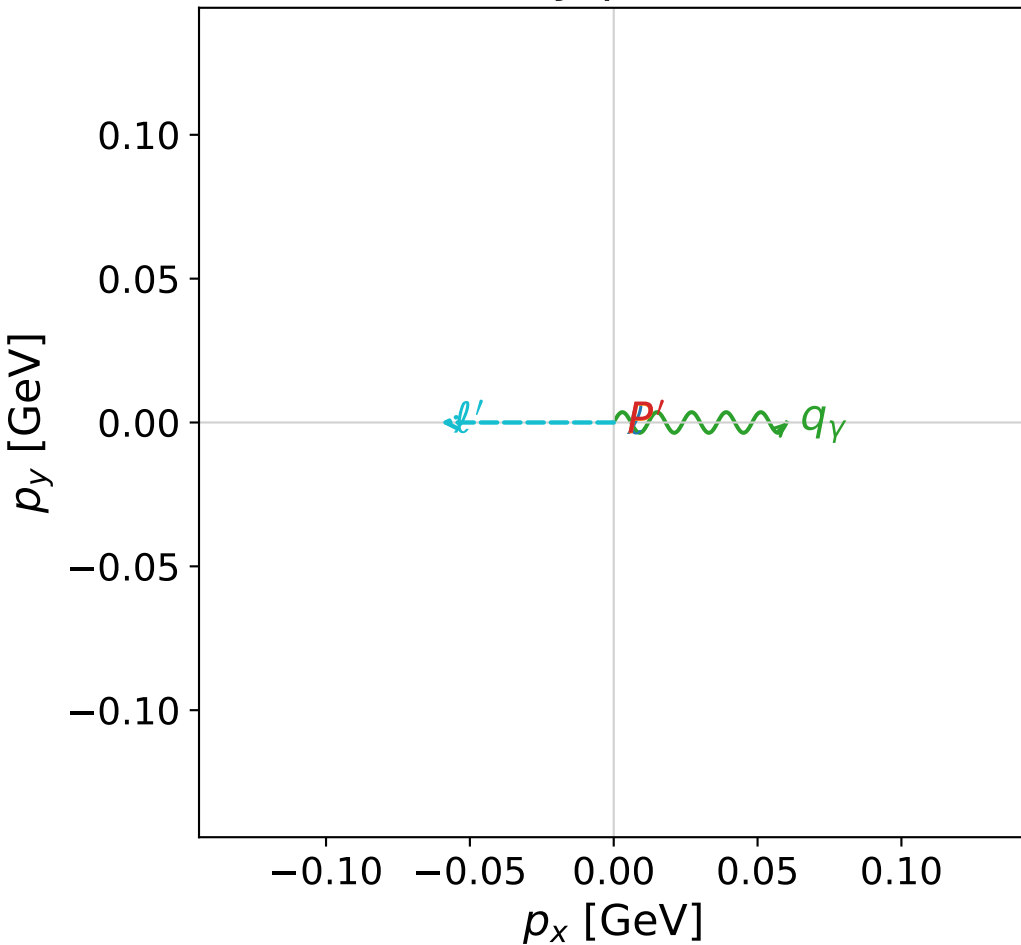


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

z--x plane



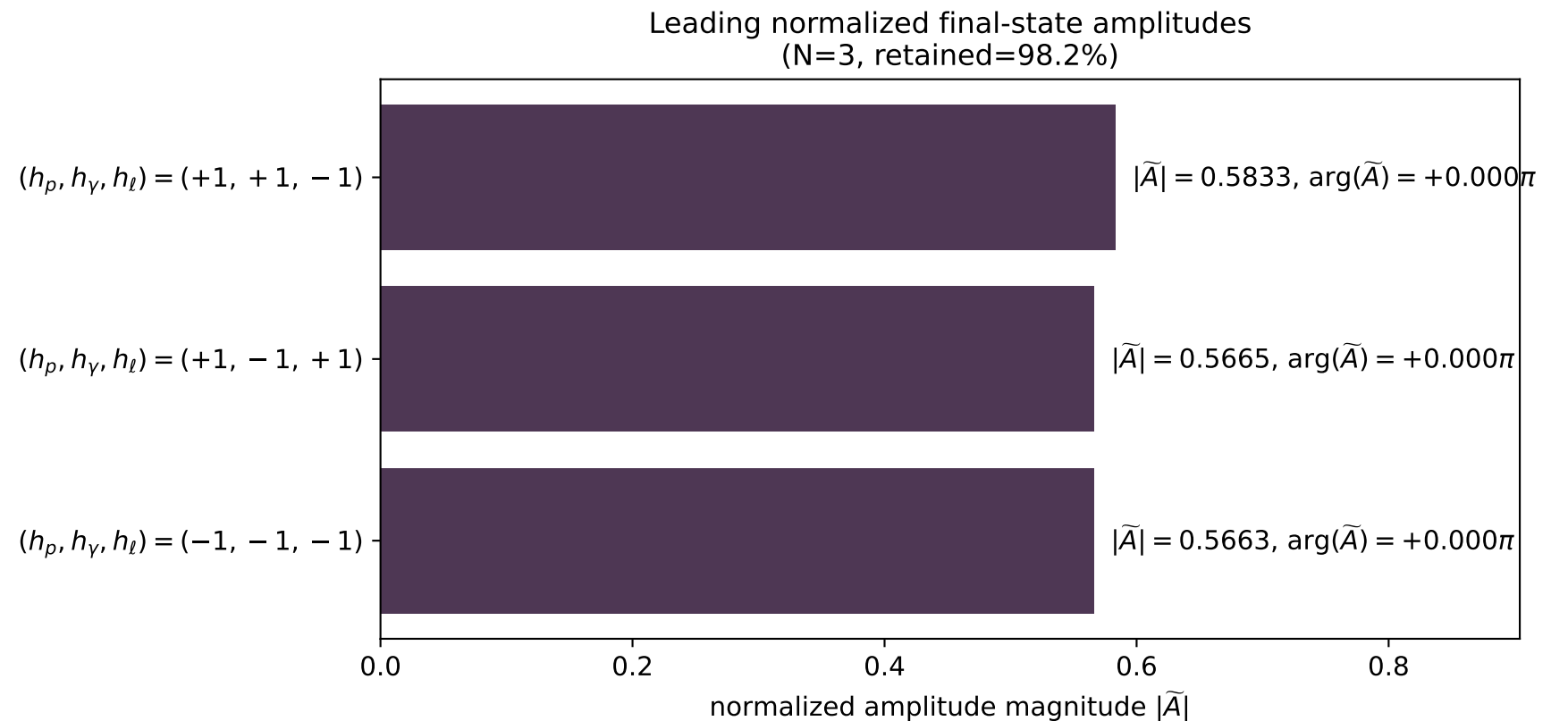
x--y plane



dw_mixing_angles_polarization_02_local_minimum_767
 $D_W=0.00946605$
 region: polarization_cluster_02_local_minimum_767
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0767
 lepton mixing angle: $\alpha_e=0.82066942$ rad
 incoming lepton: $0.681732|+\rangle + 0.731602|-\rangle$
 proton mixing angle: $\alpha_p=1.3944256$ rad
 incoming proton: $0.175458|+\rangle + 0.984487|-\rangle$

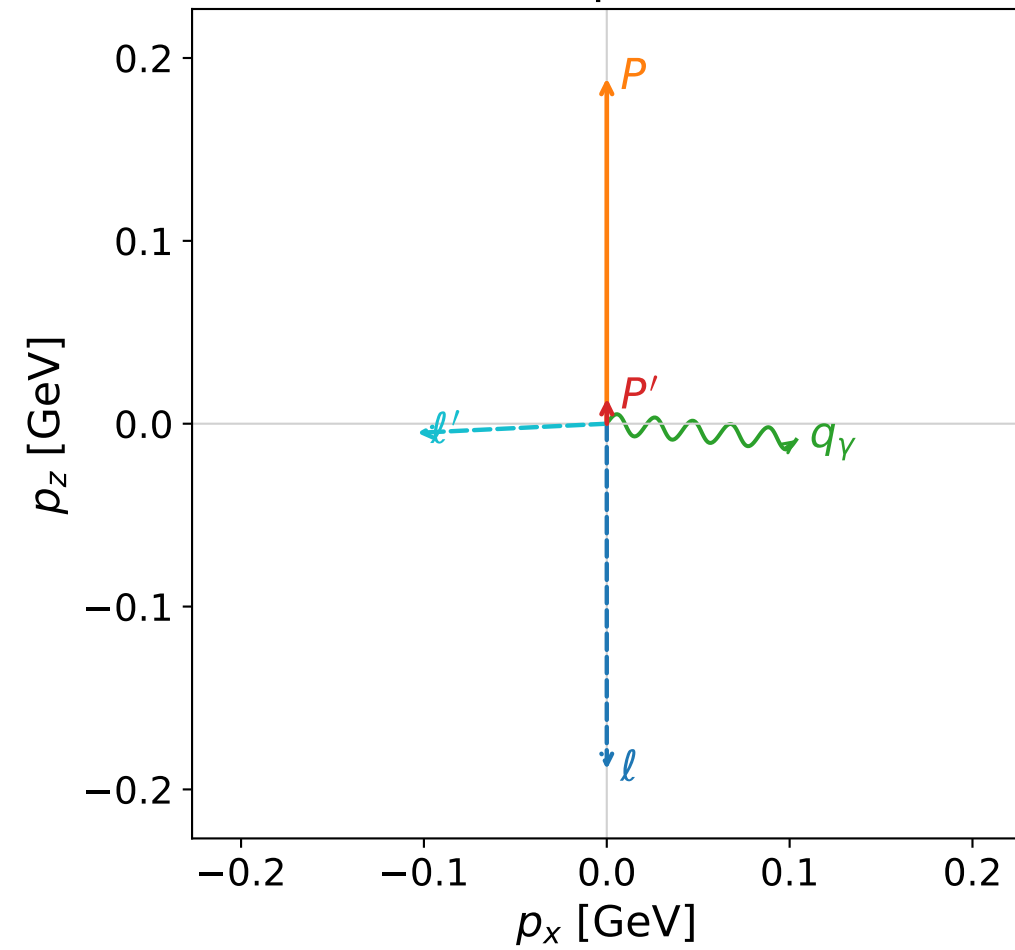
$s=1.1359$, $\sqrt{s}=1.06579$, $|\vec{P}|=0.120126$, $|\vec{P}'|=0.0396205$
 $\theta_{p'}=3.14159$, $\theta_\gamma=1.32582$, $\phi_{p'}=0$, $\phi_\gamma=0$
 $E_\gamma=0.061984$, $\theta_{l'}=1.18266$, $\phi_{l'}=3.14159$
 $Q^2=0.0215155$, $x_B=0.154685$, $t=-0.0254725$, $W^2=0.997421$, $y=0.543202$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1201$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1201$
 P $E= 0.9457$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1201$
 ℓ' $E= 0.0650$ $p_x= -0.0601$ $p_y= -0.0000$ $p_z= 0.0246$
 P' $E= 0.9388$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.0396$
 q_γ $E= 0.0620$ $p_x= 0.0601$ $p_y= 0.0000$ $p_z= 0.0150$

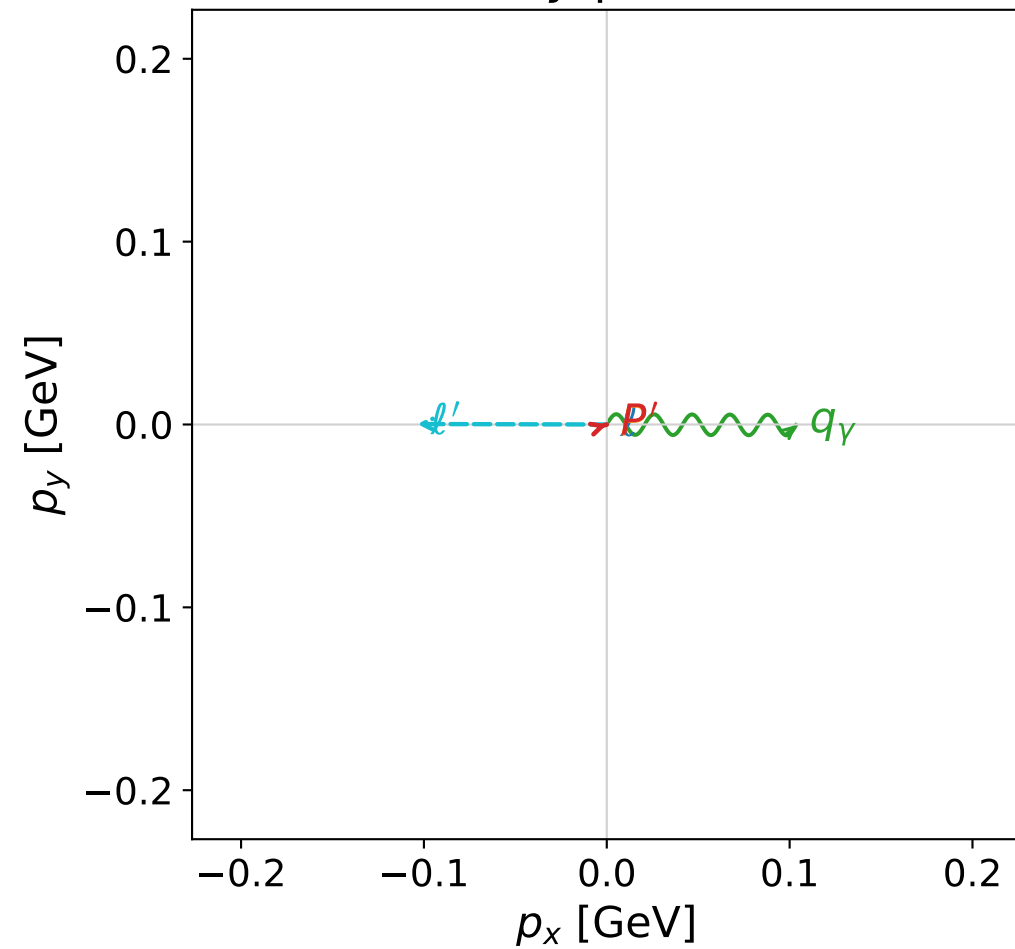


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

z--x plane



x--y plane



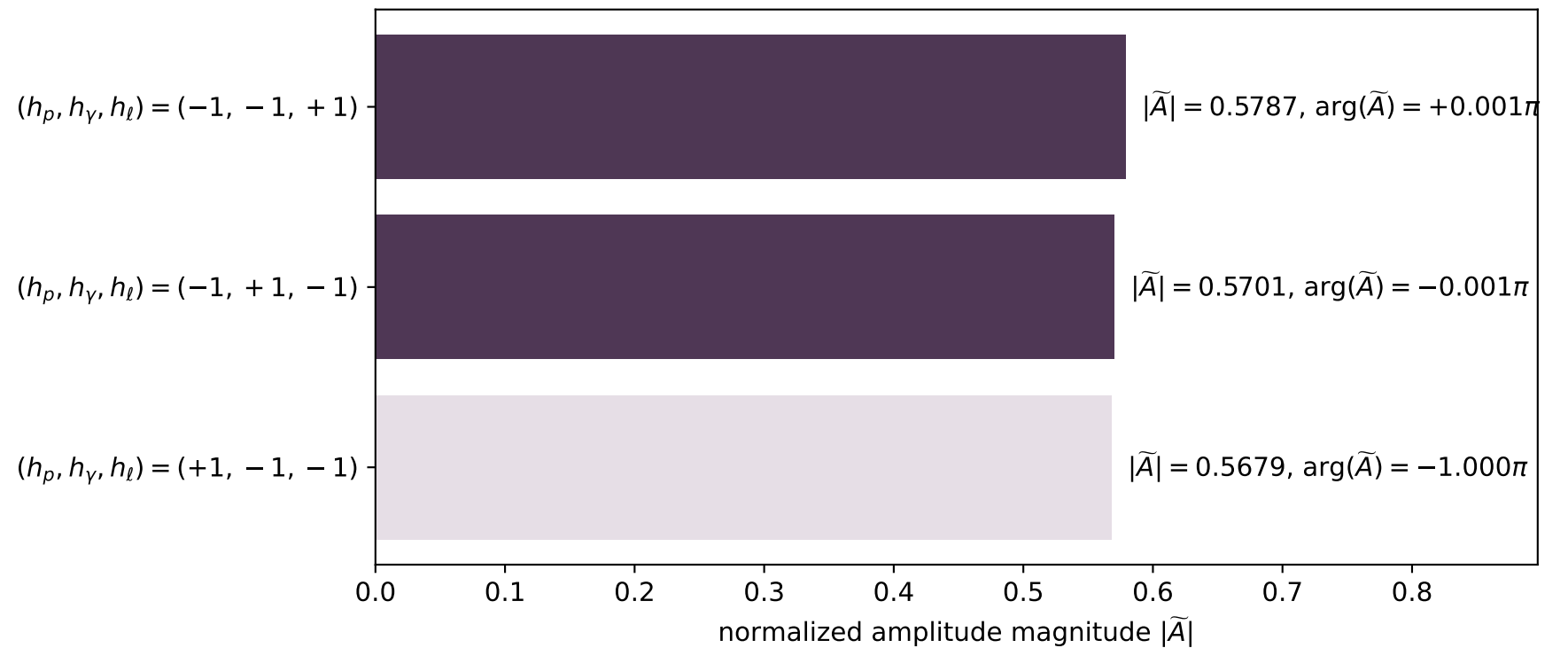
dw_mixing_angles_polarization_02_local_minimum_788
 $D_W=0.0100564$
 region: polarization_cluster_02_local_minimum_788
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0788
 lepton mixing angle: $\alpha_e=0.77394525$ rad
 incoming lepton: $0.715159|+\rangle +0.698962|-\rangle$
 proton mixing angle: $\alpha_p=1.3656881$ rad
 incoming proton: $0.203673|+\rangle +0.979039|-\rangle$

$s=1.31297$, $\sqrt{s}=1.14585$, $|\vec{P}|=0.188999$, $|\vec{P}'|=0.0138893$
 $\theta_{P'}=0.0119919$, $\theta_\gamma=1.65635$, $\phi_{P'}=0.302166$, $\phi_\gamma=6.28013$
 $E_\gamma=0.103924$, $\theta_{\ell'}=1.61905$, $\phi_{\ell'}=3.13902$
 $Q^2=0.037352$, $x_B=0.160621$, $t=-0.0303122$, $W^2=1.07504$, $y=0.536901$

four-momenta $[E, p_x, p_y, p_z]$ GeV:

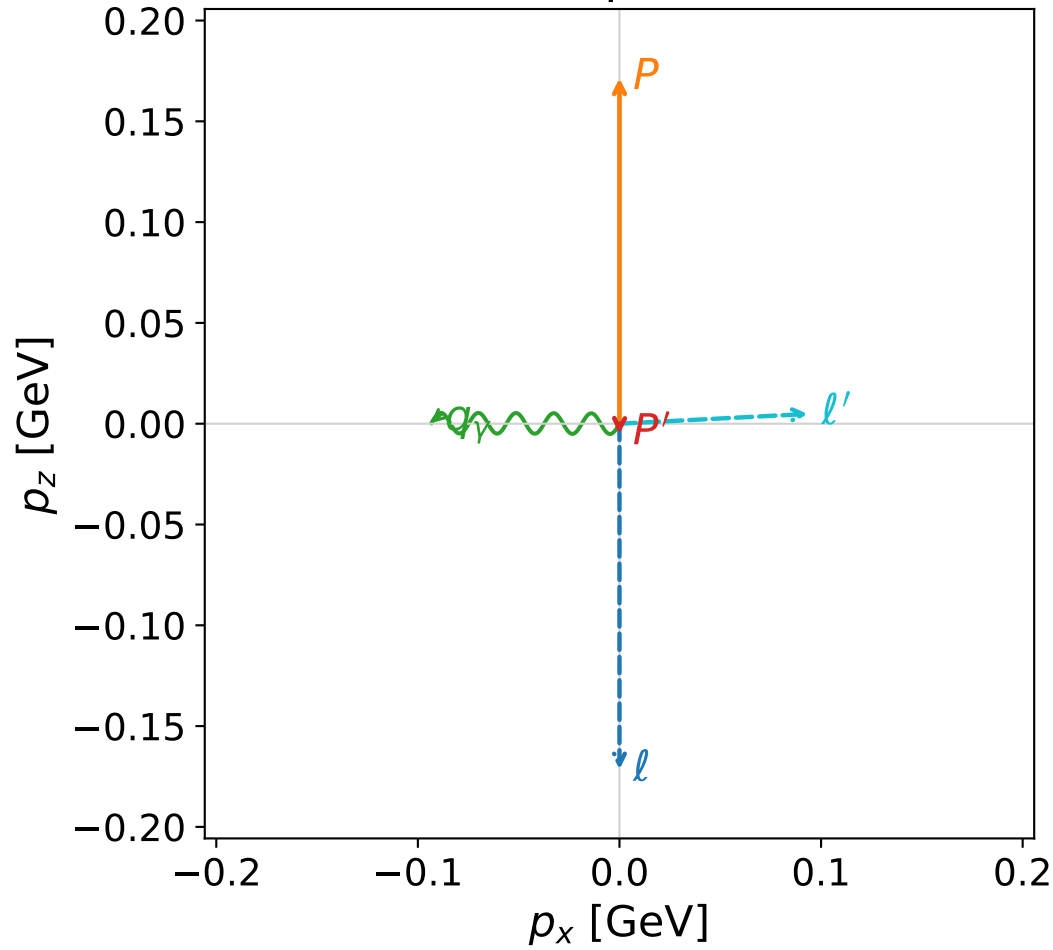
Particle	E	p_x	p_y	p_z
ℓ	0.1890	0.0000	0.0000	-0.1890
P	0.9569	0.0000	0.0000	0.1890
ℓ'	0.1038	-0.1037	0.0003	-0.0050
P'	0.9381	0.0002	0.0000	0.0139
q_γ	0.1039	0.1035	-0.0003	-0.0089

Leading normalized final-state amplitudes
(N=3, retained=98.2%)



coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

z--x plane



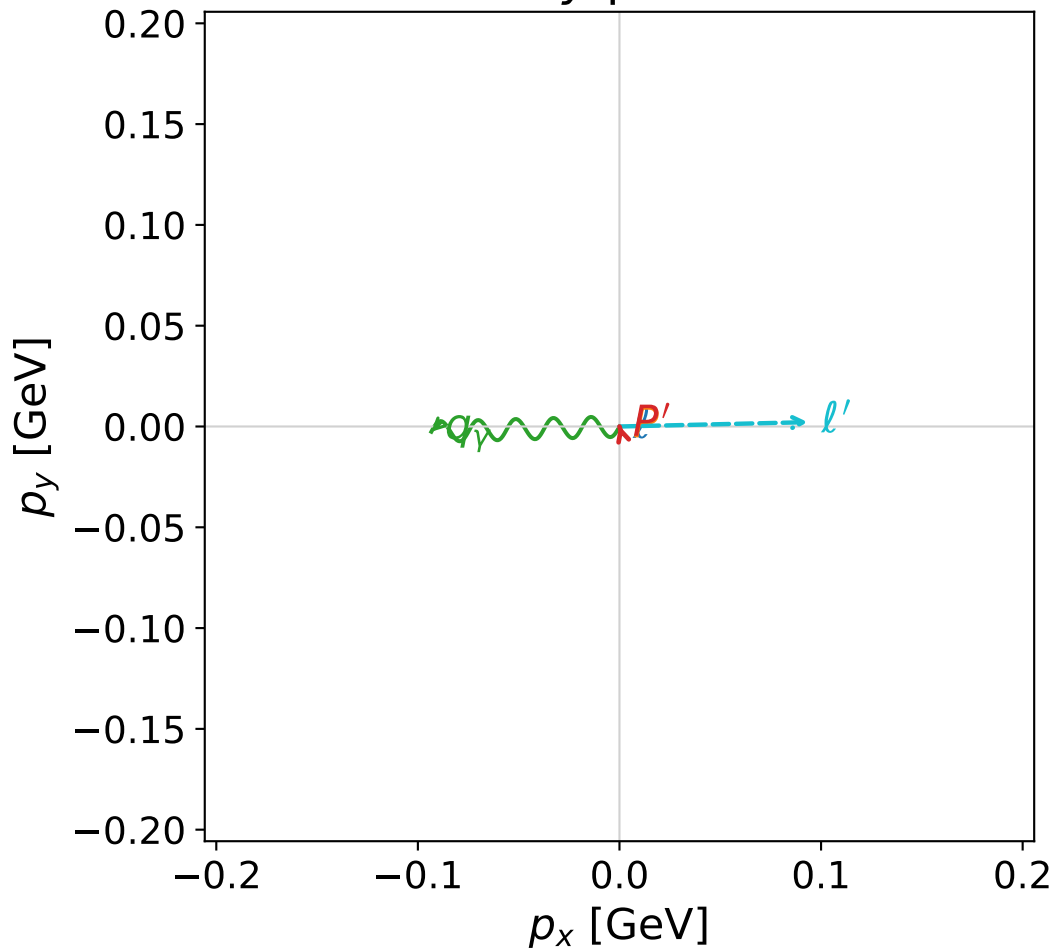
dw_mixing_angles_polarization_02_local_minimum_790
D_W=0.0101139
region: polarization_cluster_02_local_minimum_790
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0790
lepton mixing angle: $\alpha_e=0.77782162$ rad
incoming lepton: $0.712444|+\rangle + 0.701729|-\rangle$
proton mixing angle: $\alpha_p=1.7730066$ rad
incoming proton: $-0.200835|+\rangle + 0.979625|-\rangle$

$s=1.26556$, $\sqrt{s}=1.12497$, $|\vec{P}|=0.171433$, $|\vec{P}'|=0.00518528$
 $\theta_{p'}=3.05207$, $\theta_\gamma=1.56744$, $\phi_{p'}=1.85608$, $\phi_\gamma=3.16934$
 $E_\gamma=0.0933554$, $\theta_{\ell'}=1.51895$, $\phi_{\ell'}=0.0229511$
 $Q^2=0.0337555$, $x_B=0.161607$, $t=-0.0309459$, $W^2=1.05496$, $y=0.541524$

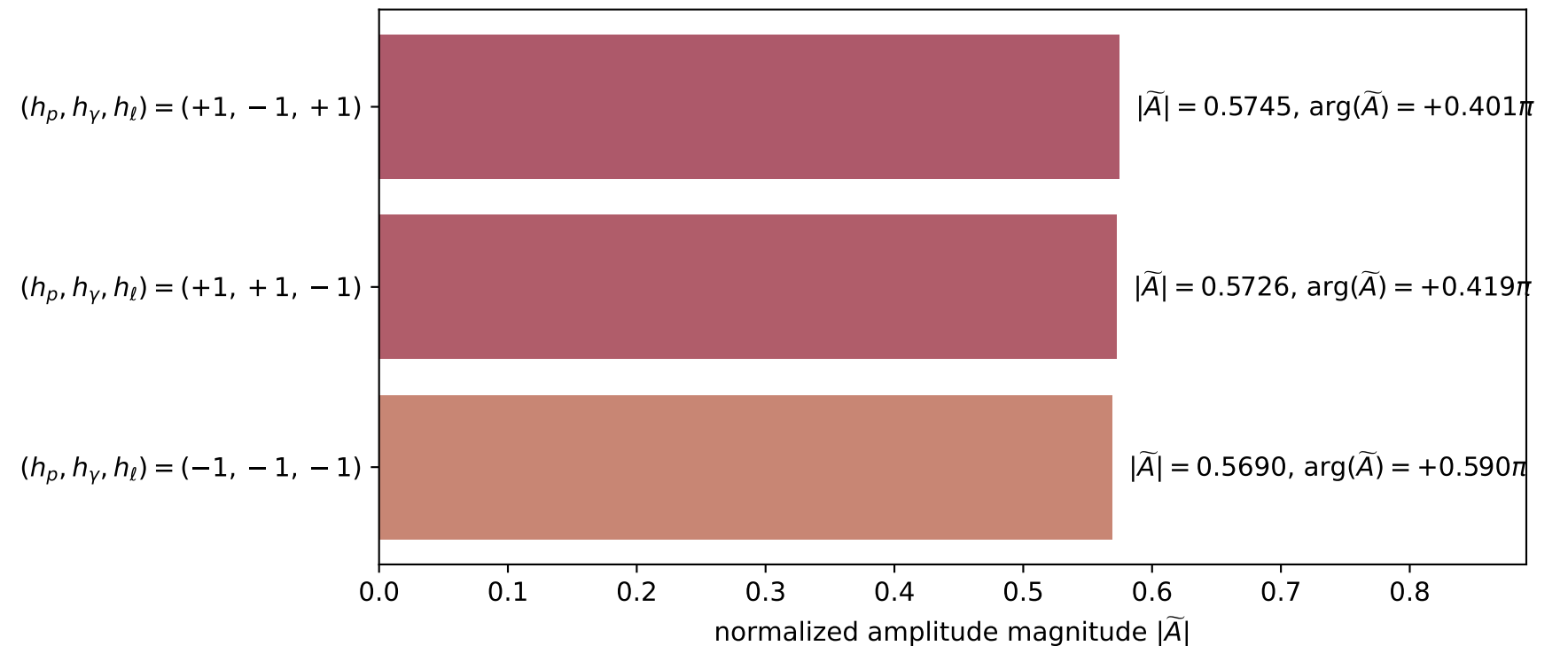
four-momenta $[E, p_x, p_y, p_z]$ GeV:

ℓ	$E=$	0.1714	$p_x=$	0.0000	$p_y=$	0.0000	$p_z=$	-0.1714
P	$E=$	0.9535	$p_x=$	0.0000	$p_y=$	0.0000	$p_z=$	0.1714
ℓ'	$E=$	0.0936	$p_x=$	0.0934	$p_y=$	0.0021	$p_z=$	0.0049
P'	$E=$	0.9380	$p_x=$	-0.0001	$p_y=$	0.0004	$p_z=$	-0.0052
q_γ	$E=$	0.0934	$p_x=$	-0.0933	$p_y=$	-0.0026	$p_z=$	0.0003

x--y plane

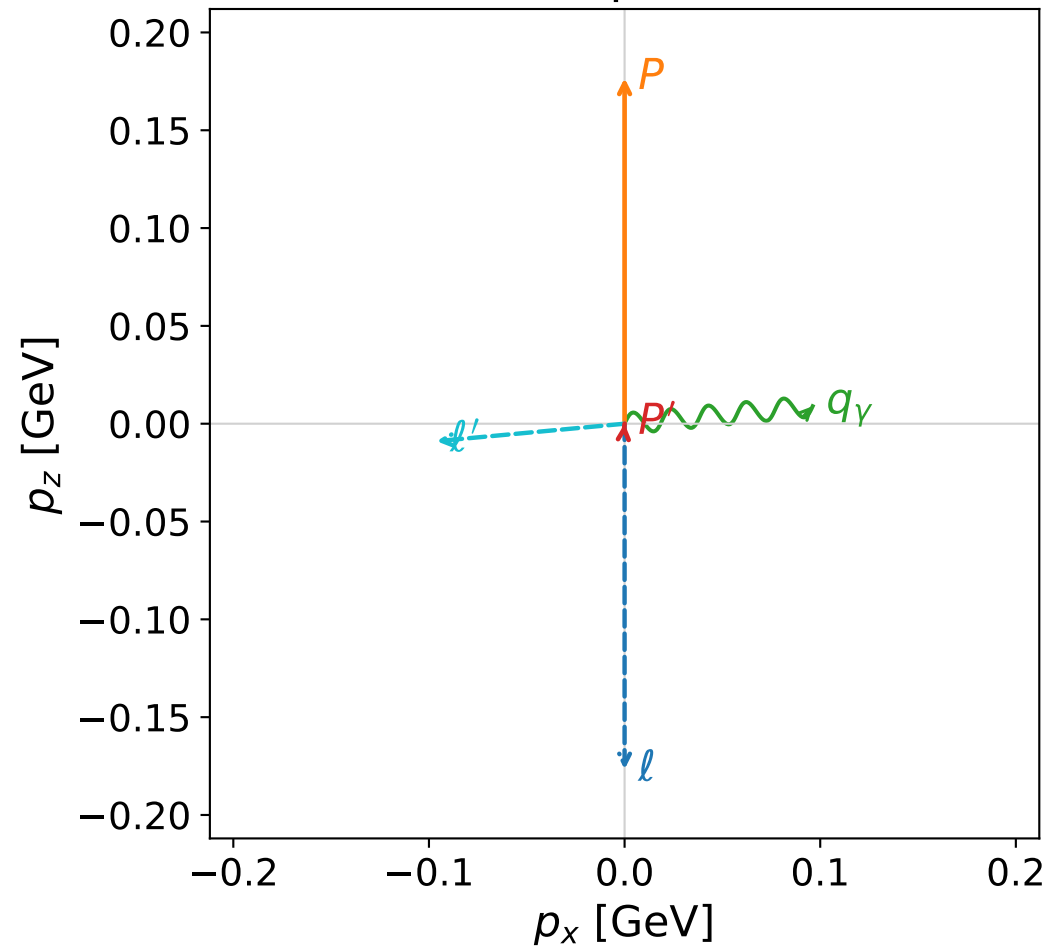


Leading normalized final-state amplitudes
(N=3, retained=98.2%)



coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

z--x plane

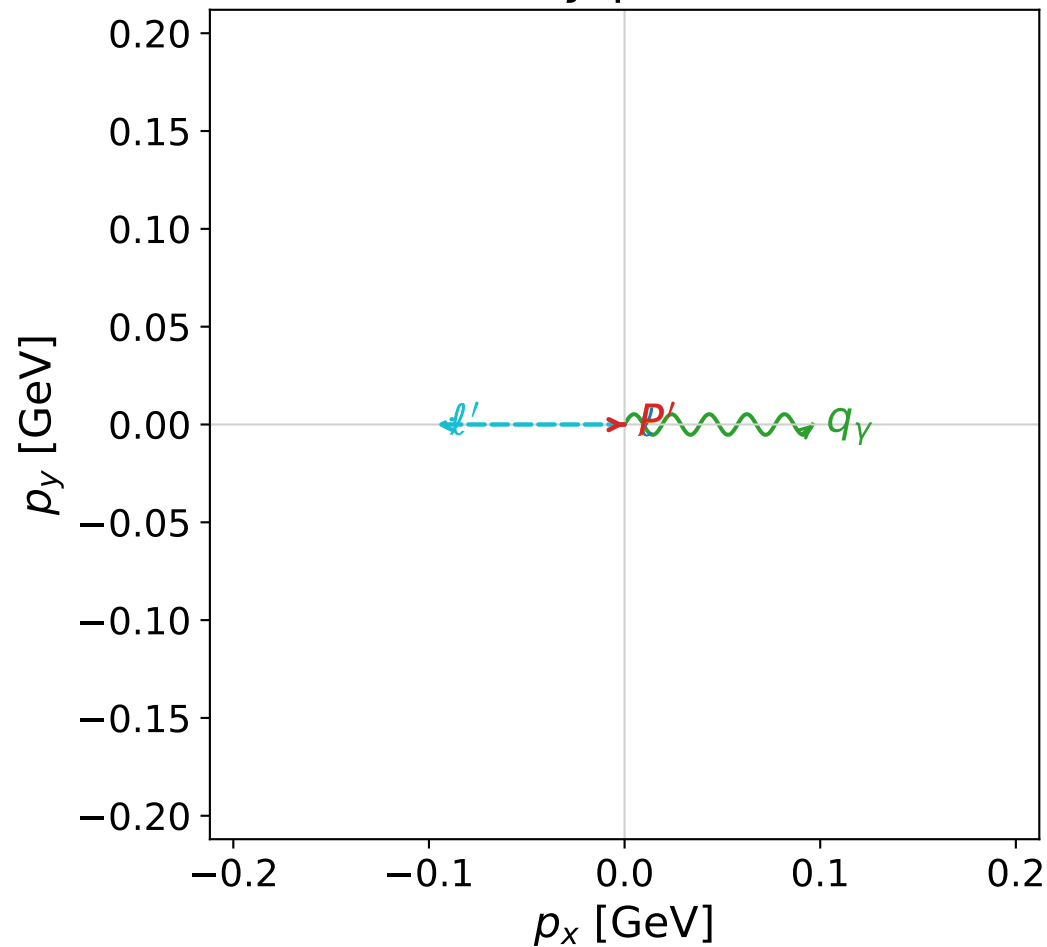


dw_mixing_angles_polarization_02_local_minimum_797
 $D_W=0.0103184$
 region: polarization_cluster_02_local_minimum_797
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0797
 lepton mixing angle: $\alpha_e=0.76992886$ rad
 incoming lepton: $0.71796|+\rangle + 0.696084|-\rangle$
 proton mixing angle: $\alpha_p=1.3669543$ rad
 incoming proton: $0.202433|+\rangle + 0.979296|-\rangle$

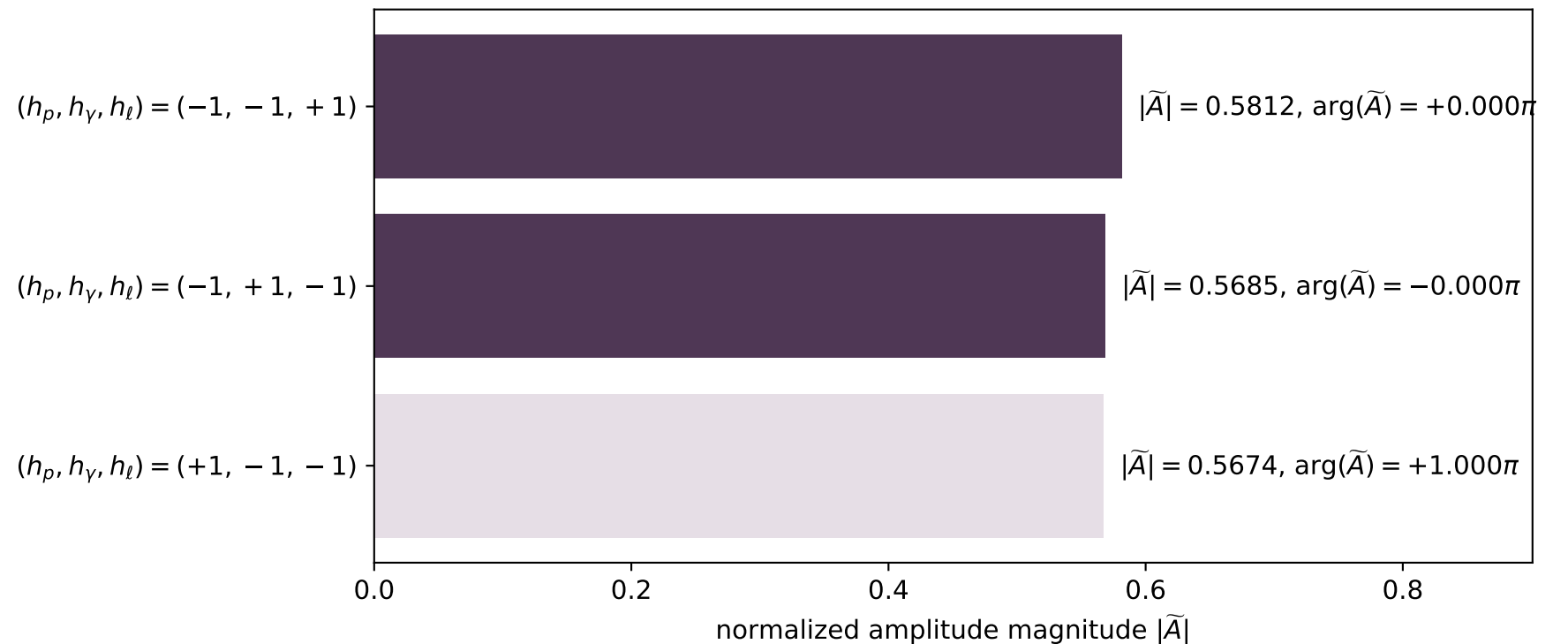
$s=1.27948$, $\sqrt{s}=1.13114$, $|\vec{P}|=0.176653$, $|\vec{P}'|=9.29299e-07$
 $\theta_{p'}=0.0778937$, $\theta_\gamma=1.47805$, $\phi_{p'}=0.00922162$, $\phi_\gamma=6.28316$
 $E_\gamma=0.0965708$, $\theta_{l'}=1.66355$, $\phi_{l'}=3.14156$
 $Q^2=0.0309589$, $x_B=0.145946$, $t=-0.030934$, $W^2=1.06101$, $y=0.530791$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E=0.1767$ $p_x=0.0000$ $p_y=0.0000$ $p_z=-0.1767$
 P $E=0.9545$ $p_x=0.0000$ $p_y=0.0000$ $p_z=0.1767$
 l' $E=0.0966$ $p_x=-0.0962$ $p_y=0.0000$ $p_z=-0.0089$
 P' $E=0.9380$ $p_x=0.0000$ $p_y=0.0000$ $p_z=0.0000$
 q_γ $E=0.0966$ $p_x=0.0962$ $p_y=-0.0000$ $p_z=0.0089$

x--y plane

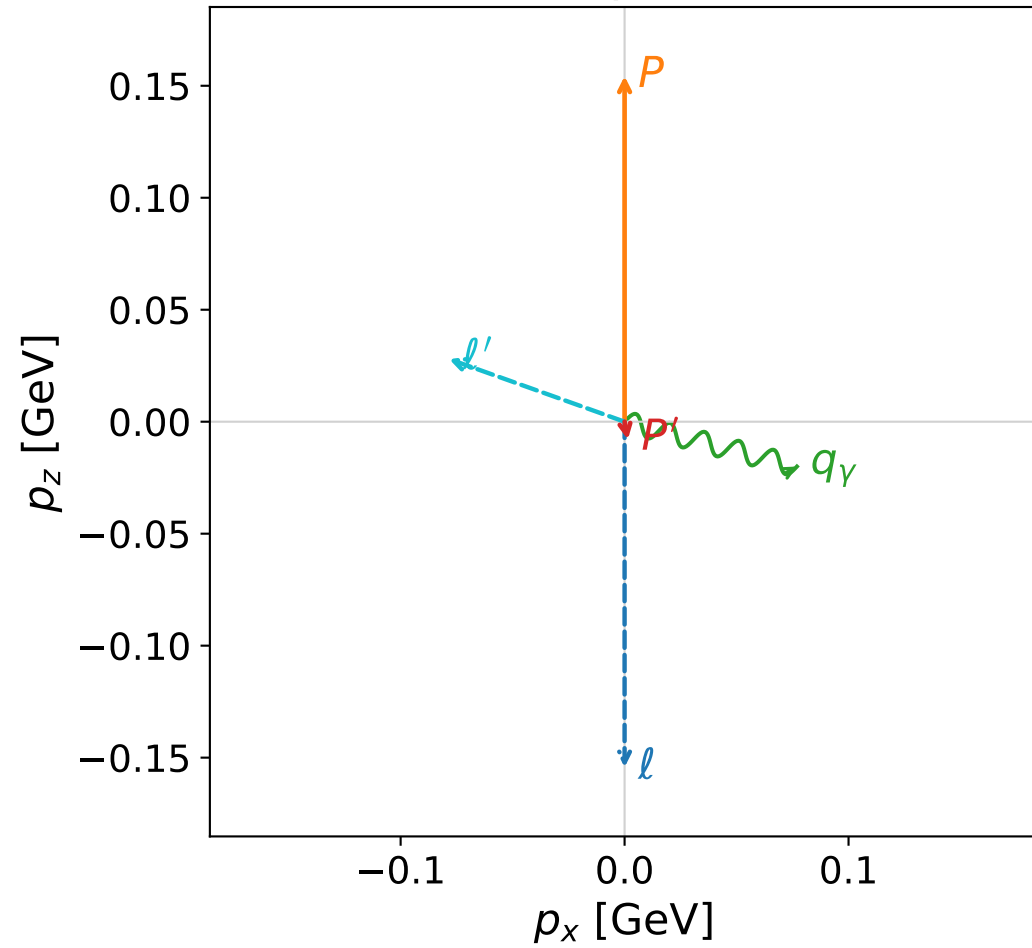


Leading normalized final-state amplitudes
 (N=3, retained=98.3%)

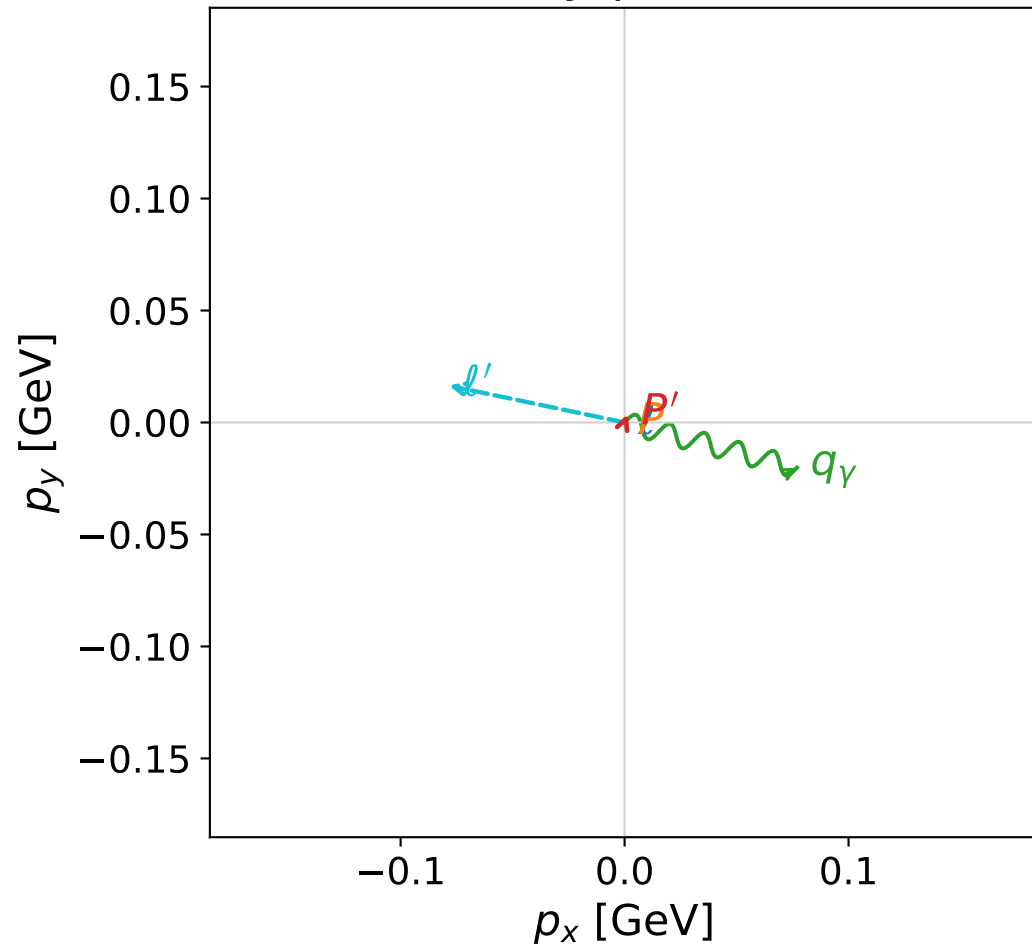


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

z--x plane



x--y plane

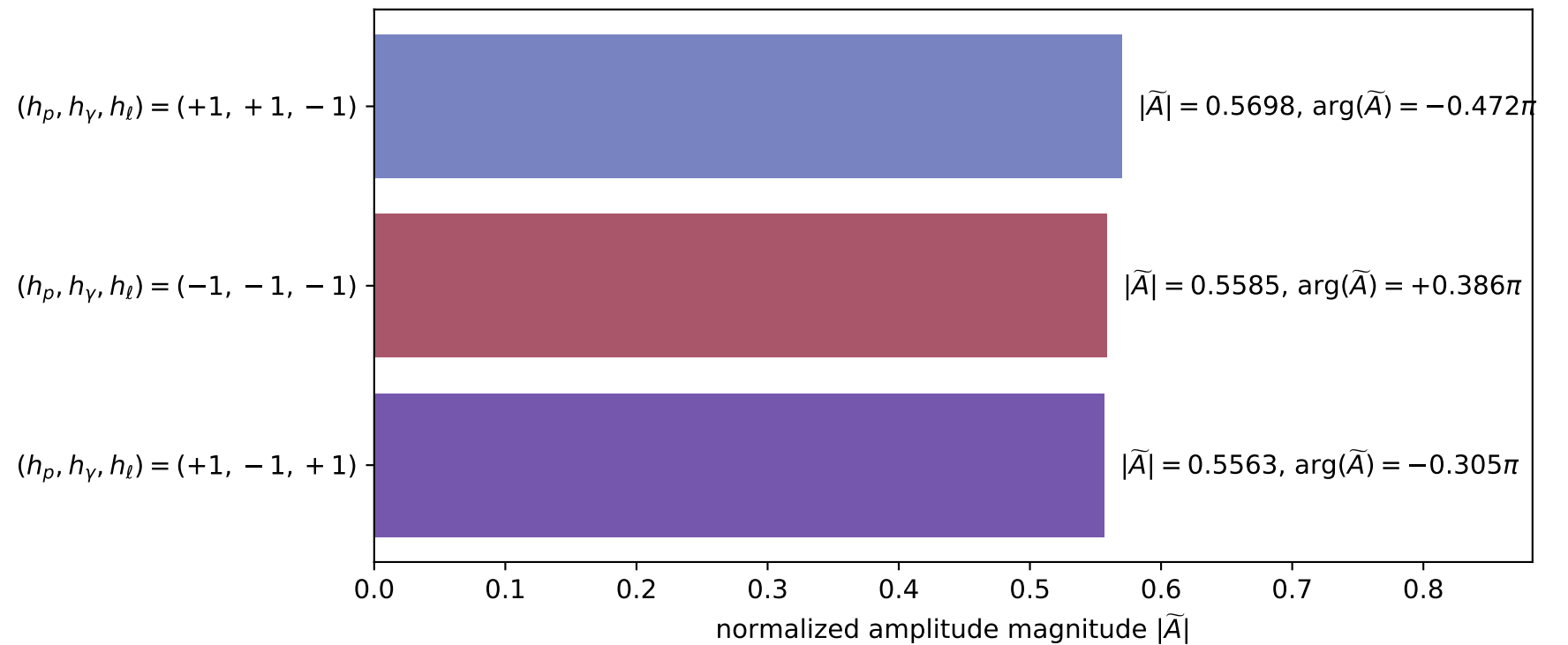


dw_mixing_angles_polarization_02_local_minimum_798
 $D_W=0.0103804$
 region: polarization_cluster_02_local_minimum_798
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0798
 lepton mixing angle: $\alpha_e=0.81788713$ rad
 incoming lepton: $0.683764|+\rangle + 0.729703|-\rangle$
 proton mixing angle: $\alpha_p=1.3602032$ rad
 incoming proton: $0.20904|+\rangle + 0.977907|-\rangle$

$s=1.22085$, $\sqrt{s}=1.10492$, $|\vec{P}|=0.154311$, $|\vec{P}'|=0.00891841$
 $\theta_{p'}=2.67511$, $\theta_\gamma=1.81721$, $\phi_{p'}=1.20325$, $\phi_\gamma=6.02686$
 $E_\gamma=0.0820325$, $\theta_{\ell'}=1.23478$, $\phi_{\ell'}=2.93505$
 $Q^2=0.0348188$, $x_B=0.184883$, $t=-0.026192$, $W^2=1.03335$, $y=0.552277$

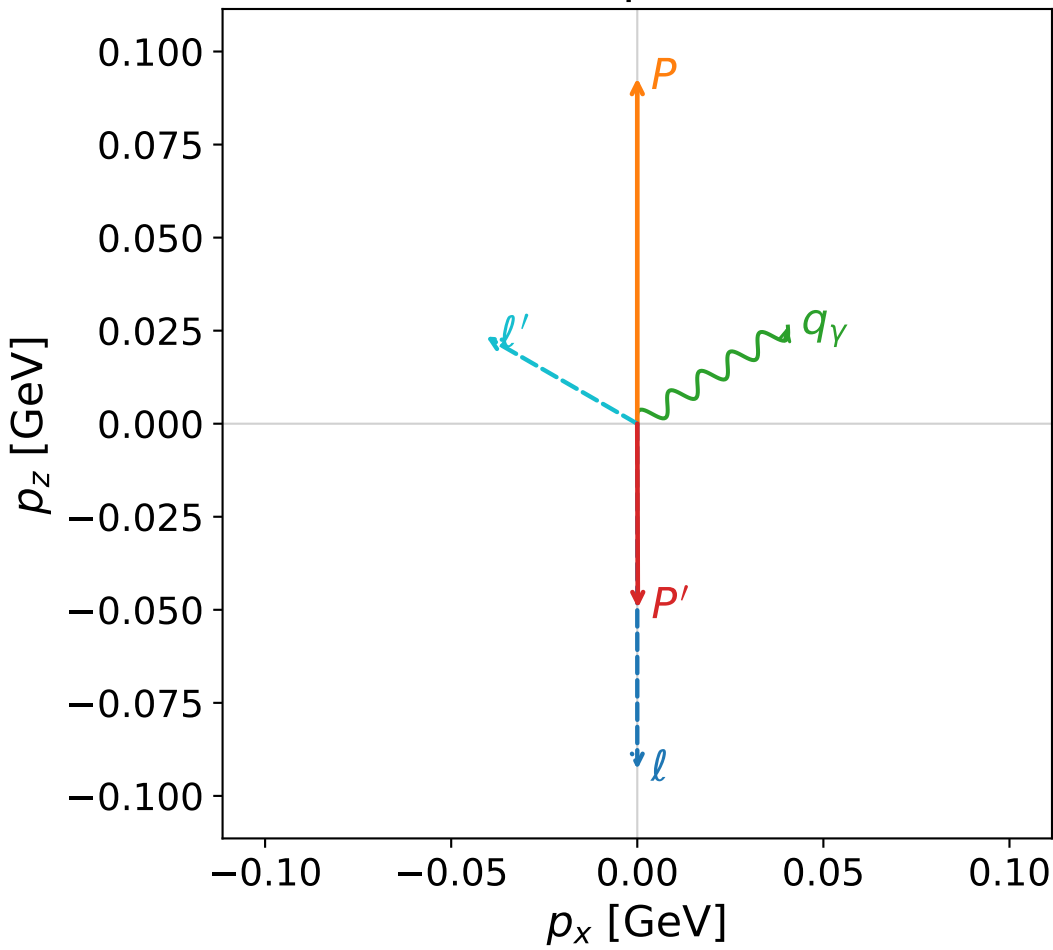
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1543$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1543$
 P $E= 0.9506$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1543$
 ℓ' $E= 0.0848$ $p_x= -0.0784$ $p_y= 0.0164$ $p_z= 0.0280$
 P' $E= 0.9380$ $p_x= 0.0014$ $p_y= 0.0037$ $p_z= -0.0080$
 q_γ $E= 0.0820$ $p_x= 0.0770$ $p_y= -0.0202$ $p_z= -0.0200$

Leading normalized final-state amplitudes
 (N=3, retained=94.6%)

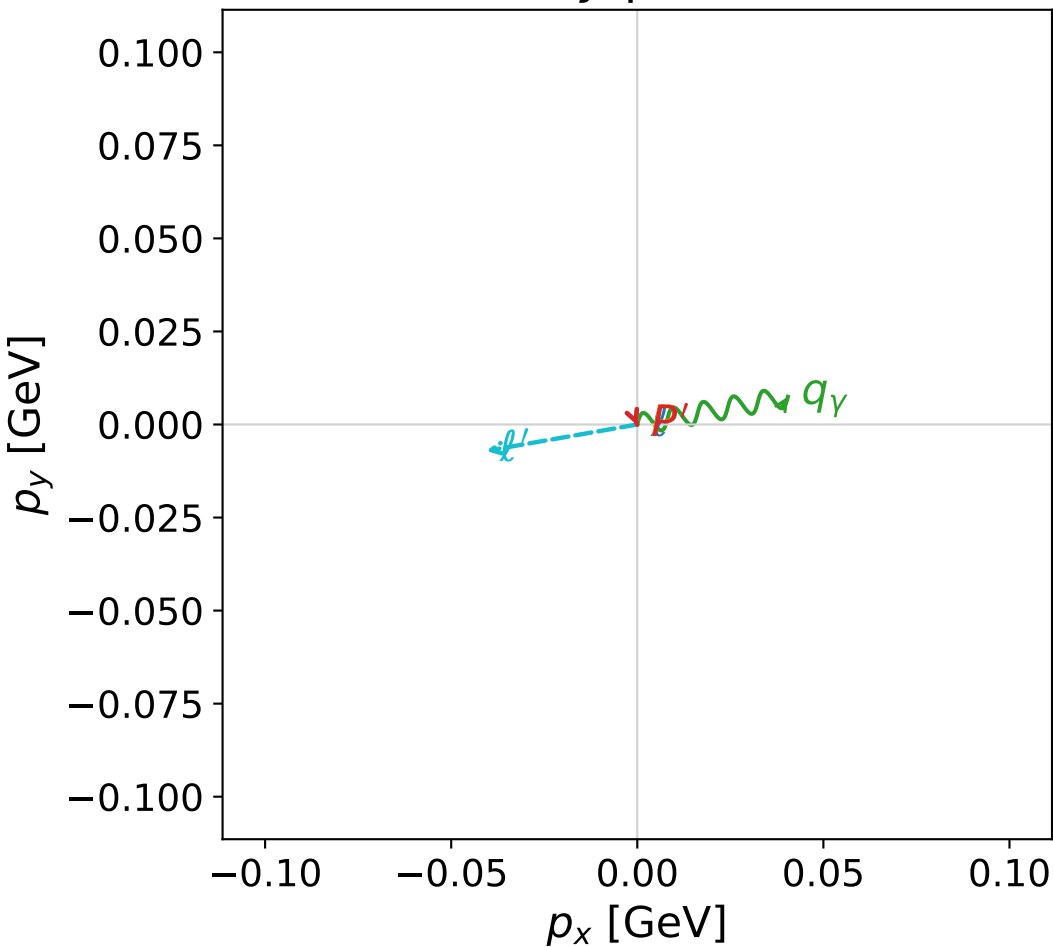


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

z--x plane



x--y plane



dw_mixing_angles_polarization_02_local_minimum_834

D_W=0.0114814

region: polarization_cluster_02_local_minimum_834

outgoing-state purity: 1

kinematic point: gradient_local_minimum_0834

lepton mixing angle: $\alpha_e=0.82148265$ rad

incoming lepton: $0.681136|+\rangle + 0.732157|-\rangle$

proton mixing angle: $\alpha_p=1.4392388$ rad

incoming proton: $0.131178|+\rangle + 0.991359|-\rangle$

$s=1.07213$, $\sqrt{s}=1.03544$, $|\vec{P}|=0.0928531$, $|\vec{P}'|=0.0495096$

$\theta_{p'}=3.13403$, $\theta_Y=1.00582$, $\phi_{p'}=5.09593$, $\phi_Y=0.182049$

$E_Y=0.0487168$, $\theta_{\ell'}=1.05404$, $\phi_{\ell'}=3.31475$

$Q^2=0.0131554$, $x_B=0.122659$, $t=-0.0202561$, $W^2=0.97394$, $y=0.557761$

four-momenta $[E, p_x, p_y, p_z]$ GeV:

ℓ $E= 0.0929$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.0929$

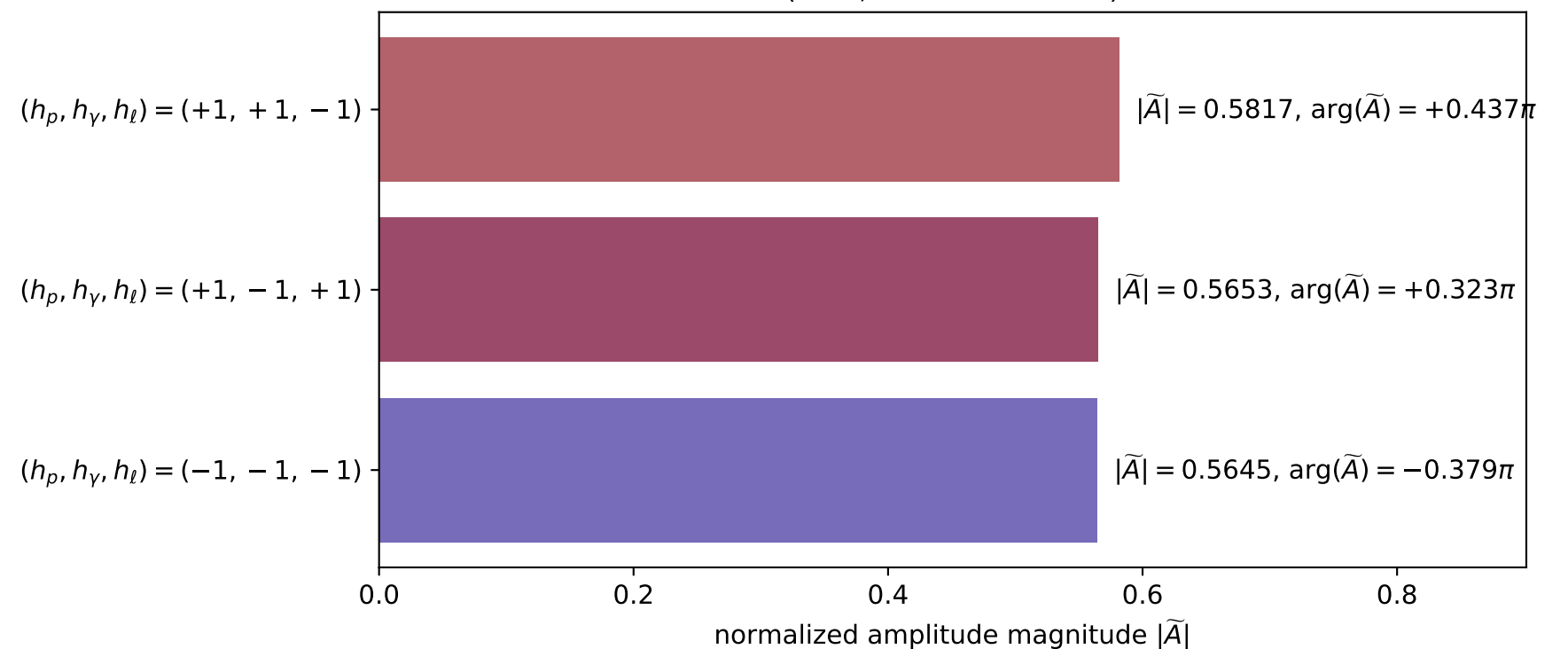
P $E= 0.9426$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.0929$

ℓ' $E= 0.0474$ $p_x= -0.0406$ $p_y= -0.0071$ $p_z= 0.0234$

P' $E= 0.9393$ $p_x= 0.0001$ $p_y= -0.0003$ $p_z= -0.0495$

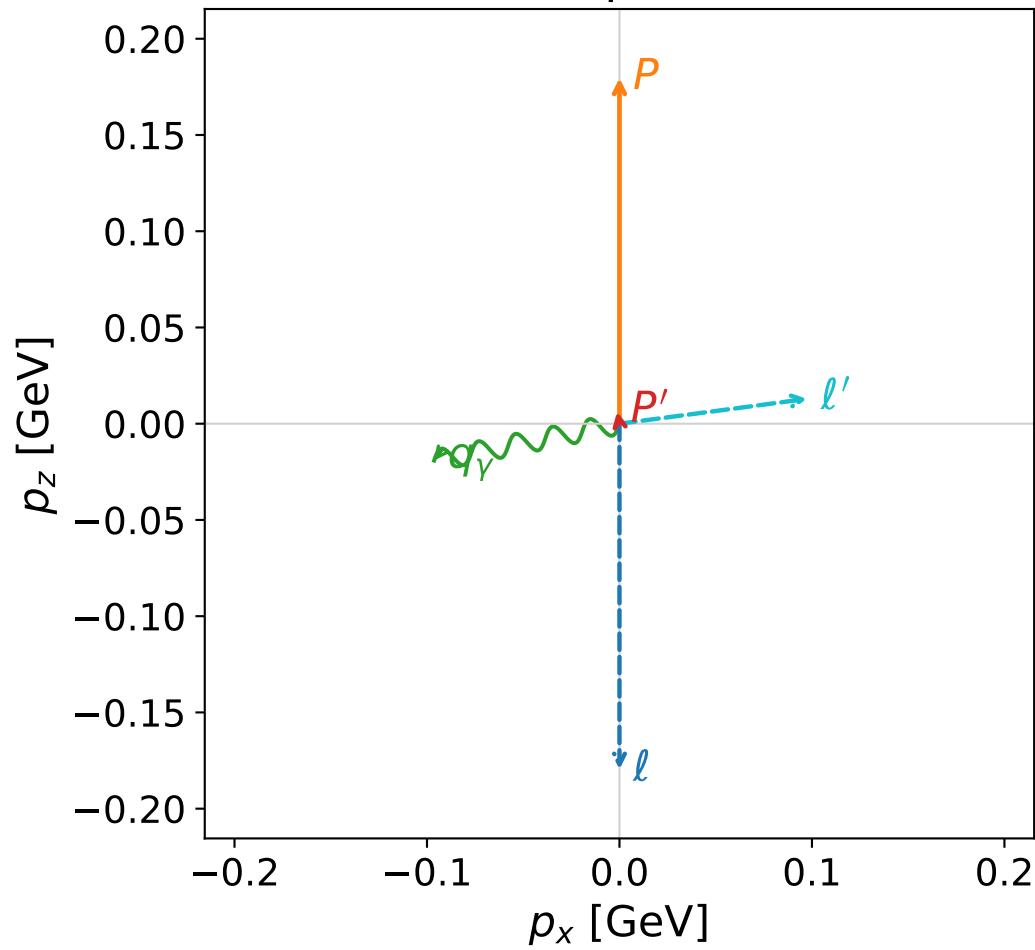
q_Y $E= 0.0487$ $p_x= 0.0405$ $p_y= 0.0074$ $p_z= 0.0261$

Leading normalized final-state amplitudes
(N=3, retained=97.7%)



coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

z--x plane



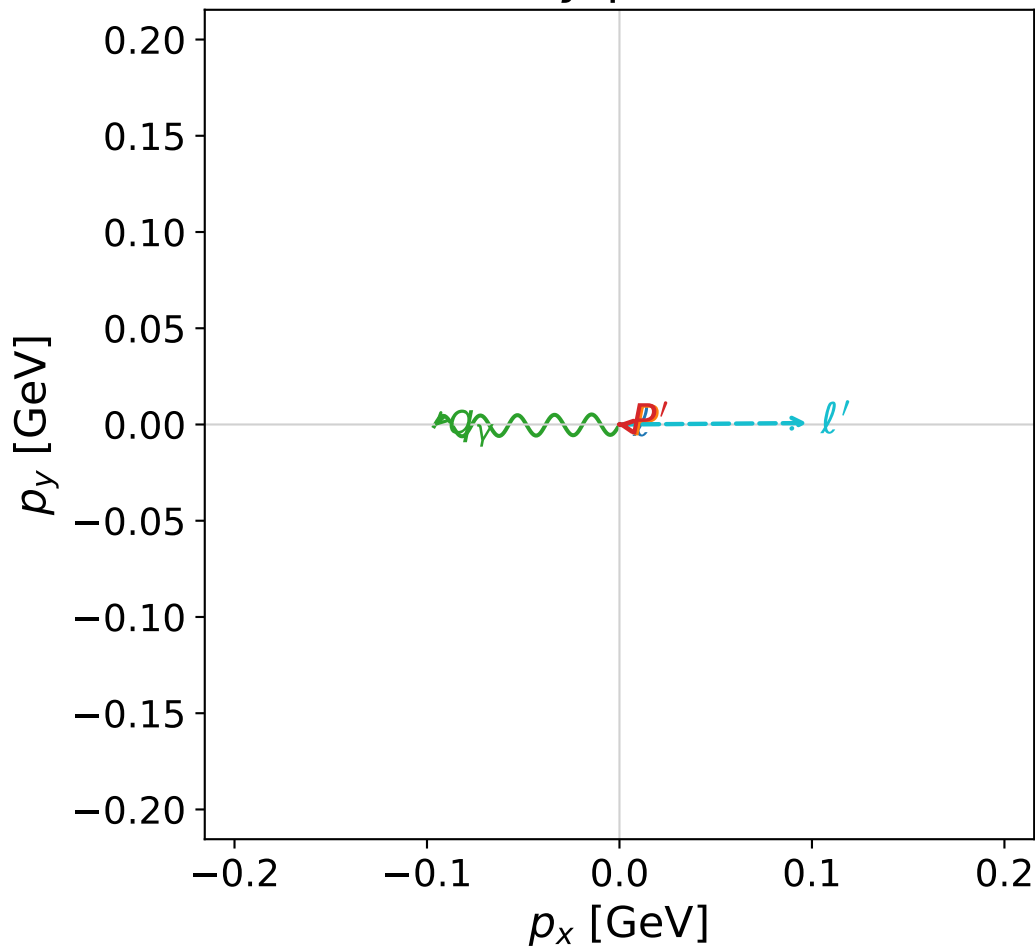
dw_mixing_angles_polarization_02_local_minimum_837
 $D_W=0.0115901$
 region: polarization_cluster_02_local_minimum_837
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0837
 lepton mixing angle: $\alpha_e=0.78523567$ rad
 incoming lepton: $0.707222|+\rangle + 0.706992|-\rangle$
 proton mixing angle: $\alpha_p=1.7803765$ rad
 incoming proton: $-0.208049|+\rangle + 0.978118|-\rangle$

$s=1.28726$, $\sqrt{s}=1.13458$, $|\vec{P}|=0.179546$, $|\vec{P}'|=0.00639123$
 $\theta_{p'}=0.190187$, $\theta_\gamma=1.76725$, $\phi_{p'}=3.00613$, $\phi_\gamma=3.15113$
 $E_\gamma=0.0981989$, $\theta_{l'}=1.43934$, $\phi_{l'}=0.00774615$
 $Q^2=0.0399477$, $x_B=0.178192$, $t=-0.0297348$, $W^2=1.06408$, $y=0.550253$

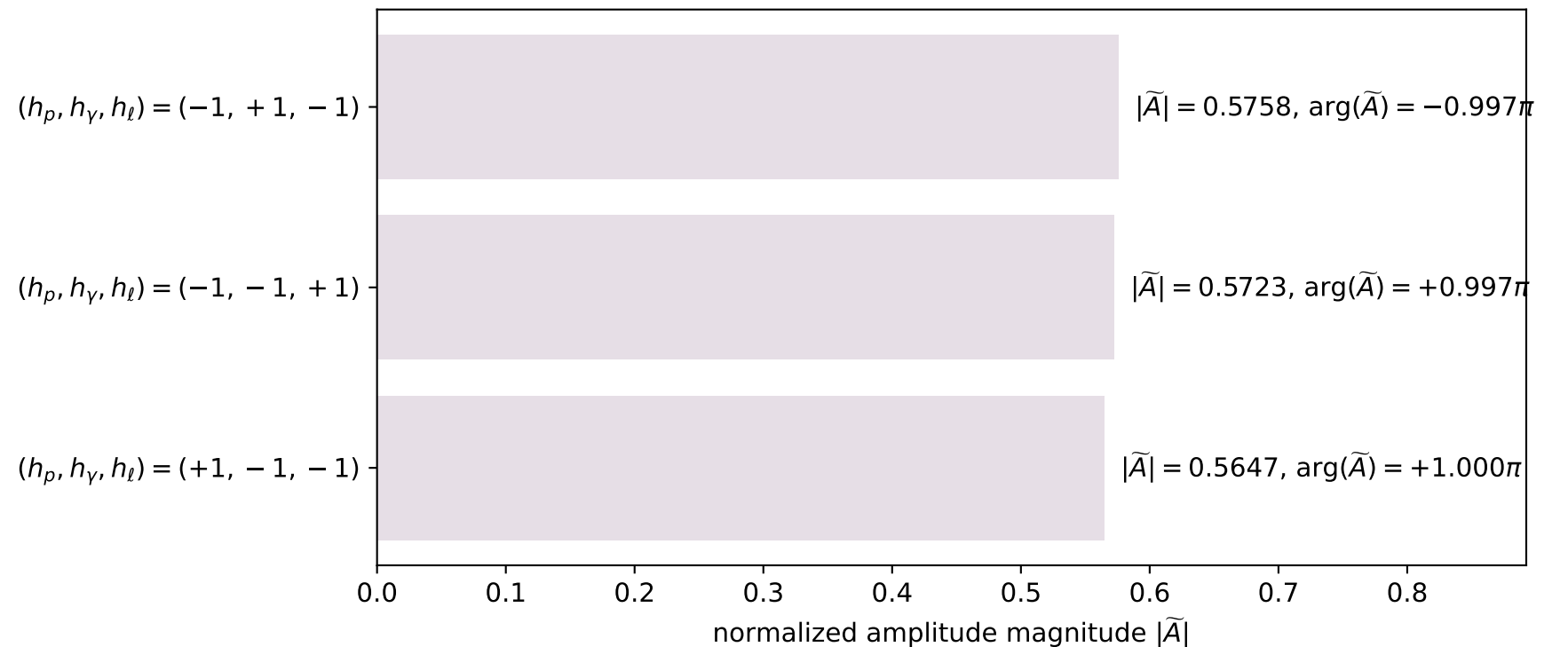
four-momenta $[E, p_x, p_y, p_z]$ GeV:

l	$E=$	0.1795	$p_x=$	0.0000	$p_y=$	0.0000	$p_z=$	-0.1795
P	$E=$	0.9550	$p_x=$	0.0000	$p_y=$	0.0000	$p_z=$	0.1795
l'	$E=$	0.0984	$p_x=$	0.0975	$p_y=$	0.0008	$p_z=$	0.0129
P'	$E=$	0.9380	$p_x=$	-0.0012	$p_y=$	0.0002	$p_z=$	0.0063
q_γ	$E=$	0.0982	$p_x=$	-0.0963	$p_y=$	-0.0009	$p_z=$	-0.0192

x--y plane

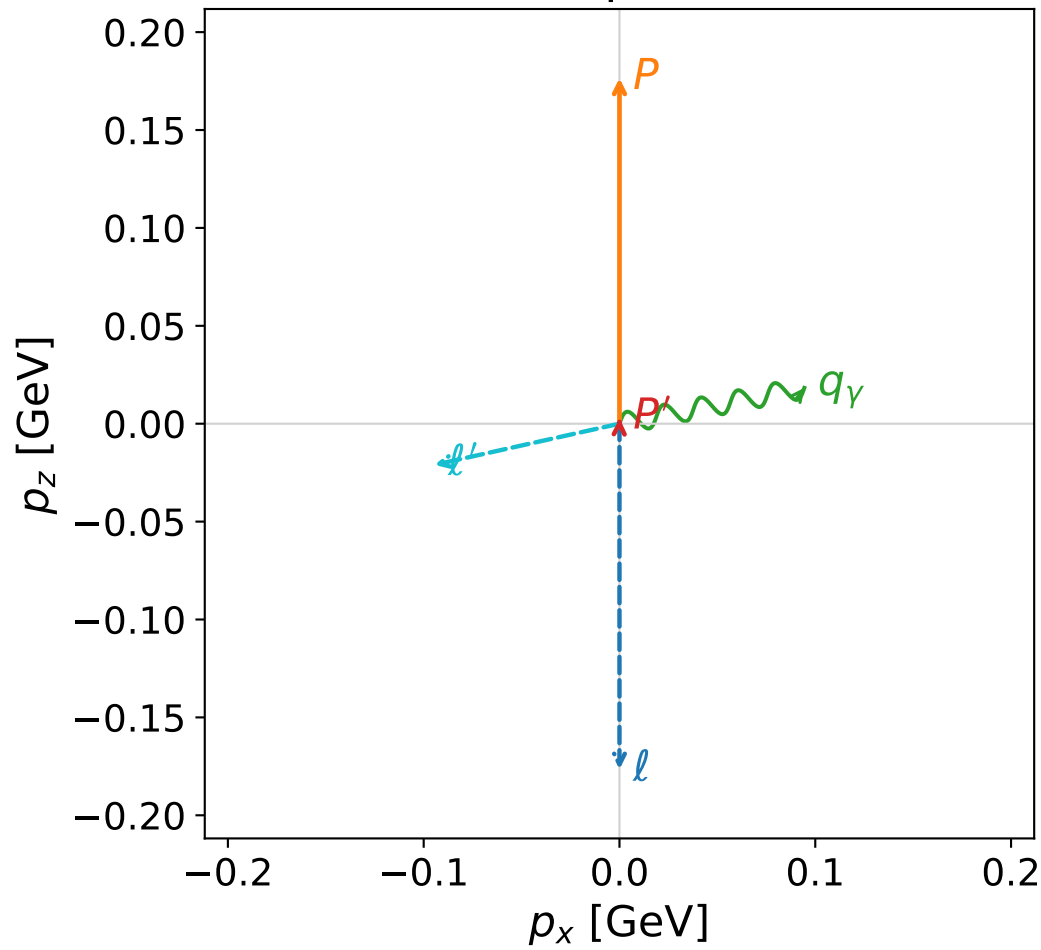


Leading normalized final-state amplitudes
(N=3, retained=97.8%)

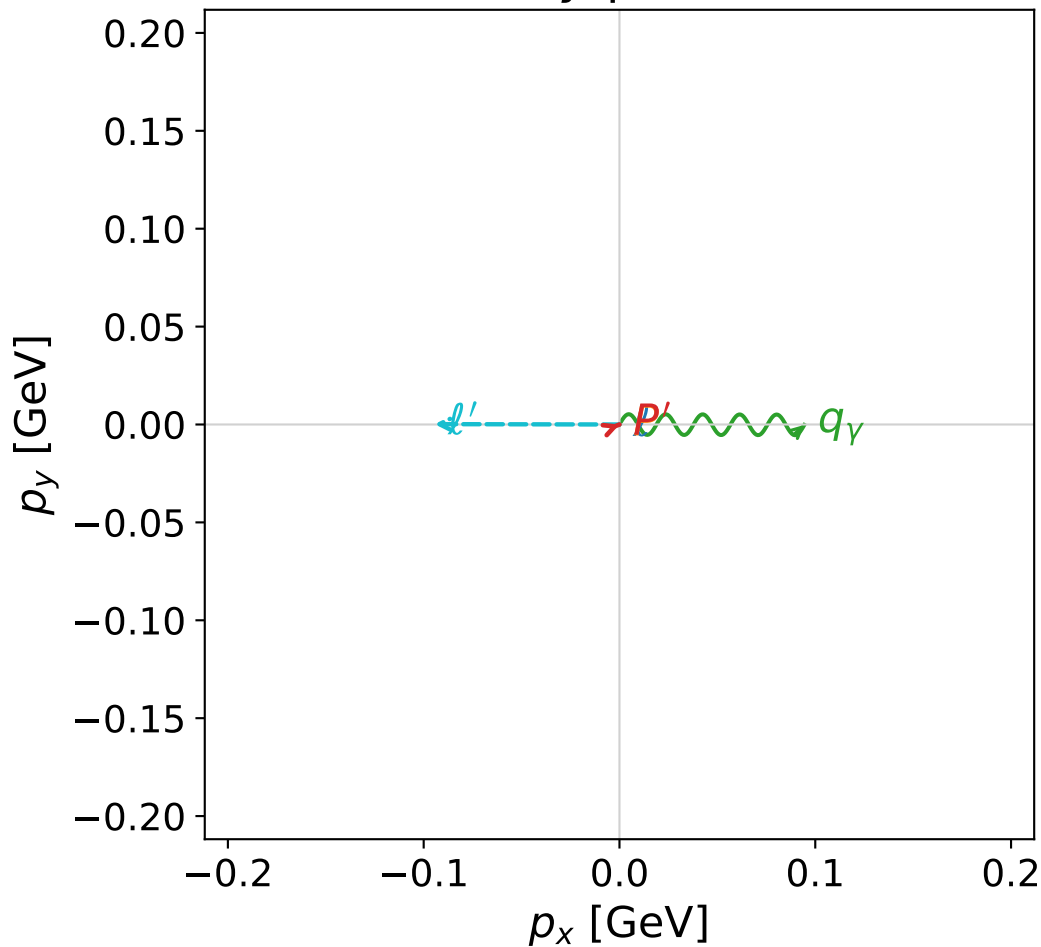


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

z--x plane



x--y plane



dw_mixing_angles_polarization_02_local_minimum_872

D_W=0.0124982

region: polarization_cluster_02_local_minimum_872

outgoing-state purity: 1

kinematic point: gradient_local_minimum_0872

lepton mixing angle: $\alpha_e=0.75872549$ rad

incoming lepton: $0.725713|+\rangle + 0.687997|-\rangle$

proton mixing angle: $\alpha_p=1.3797863$ rad

incoming proton: $0.189851|+\rangle + 0.981813|-\rangle$

$s=1.2791$, $\sqrt{s}=1.13098$, $|\vec{P}|=0.176511$, $|\vec{P}'|=0.00273129$

$\theta_{p'}=0.0285799$, $\theta_Y=1.37942$, $\phi_{p'}=0.367067$, $\phi_Y=6.28093$

$E_Y=0.0961711$, $\theta_{\ell'}=1.78973$, $\phi_{\ell'}=3.13964$

$Q^2=0.0267504$, $x_B=0.129195$, $t=-0.0299289$, $W^2=1.06015$, $y=0.518595$

four-momenta $[E, p_x, p_y, p_z]$ GeV:

ℓ $E= 0.1765$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1765$

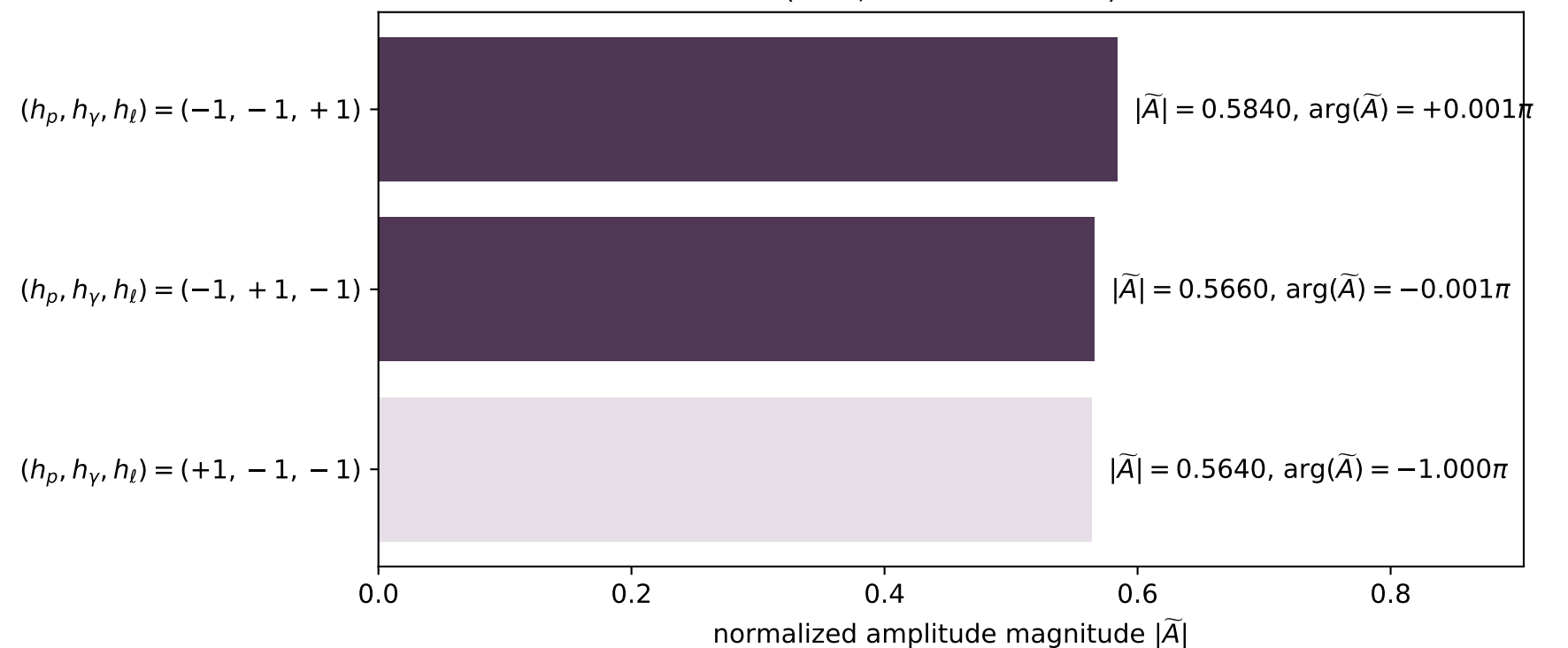
P $E= 0.9545$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1765$

ℓ' $E= 0.0968$ $p_x= -0.0945$ $p_y= 0.0002$ $p_z= -0.0210$

P' $E= 0.9380$ $p_x= 0.0001$ $p_y= 0.0000$ $p_z= 0.0027$

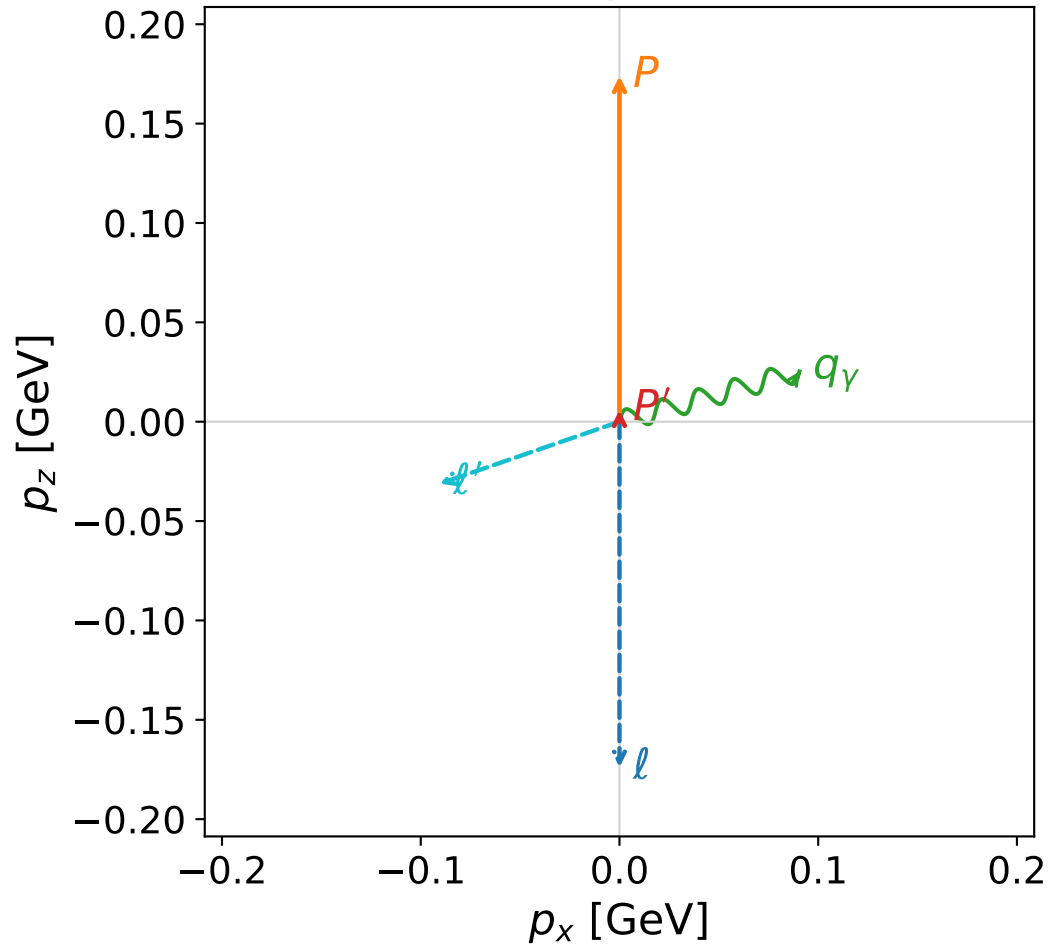
q_Y $E= 0.0962$ $p_x= 0.0944$ $p_y= -0.0002$ $p_z= 0.0183$

Leading normalized final-state amplitudes
(N=3, retained=97.9%)



coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

z--x plane



dw_mixing_angles_polarization_02_local_minimum_928

D_W=0.0148606

region: polarization_cluster_02_local_minimum_928

outgoing-state purity: 1

kinematic point: gradient_local_minimum_0928

lepton mixing angle: $\alpha_e=0.75409916$ rad

incoming lepton: $0.728889|+\rangle + 0.684632|-\rangle$

proton mixing angle: $\alpha_p=1.4076007$ rad

incoming proton: $0.162472|+\rangle + 0.986713|-\rangle$

$s=1.27212$, $\sqrt{s}=1.12788$, $|\vec{P}|=0.173898$, $|\vec{P}'|=0.00591441$

$\theta_{p'}=0$, $\theta_\gamma=1.2983$, $\phi_{p'}=0.768168$, $\phi_\gamma=2.86224e-05$

$E_\gamma=0.0940503$, $\theta_{\ell'}=1.90278$, $\phi_{\ell'}=3.14162$

$Q^2=0.0224623$, $x_B=0.113101$, $t=-0.0279635$, $W^2=1.05599$, $y=0.506292$

four-momenta $[E, p_x, p_y, p_z]$ GeV:

ℓ $E= 0.1739$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1739$

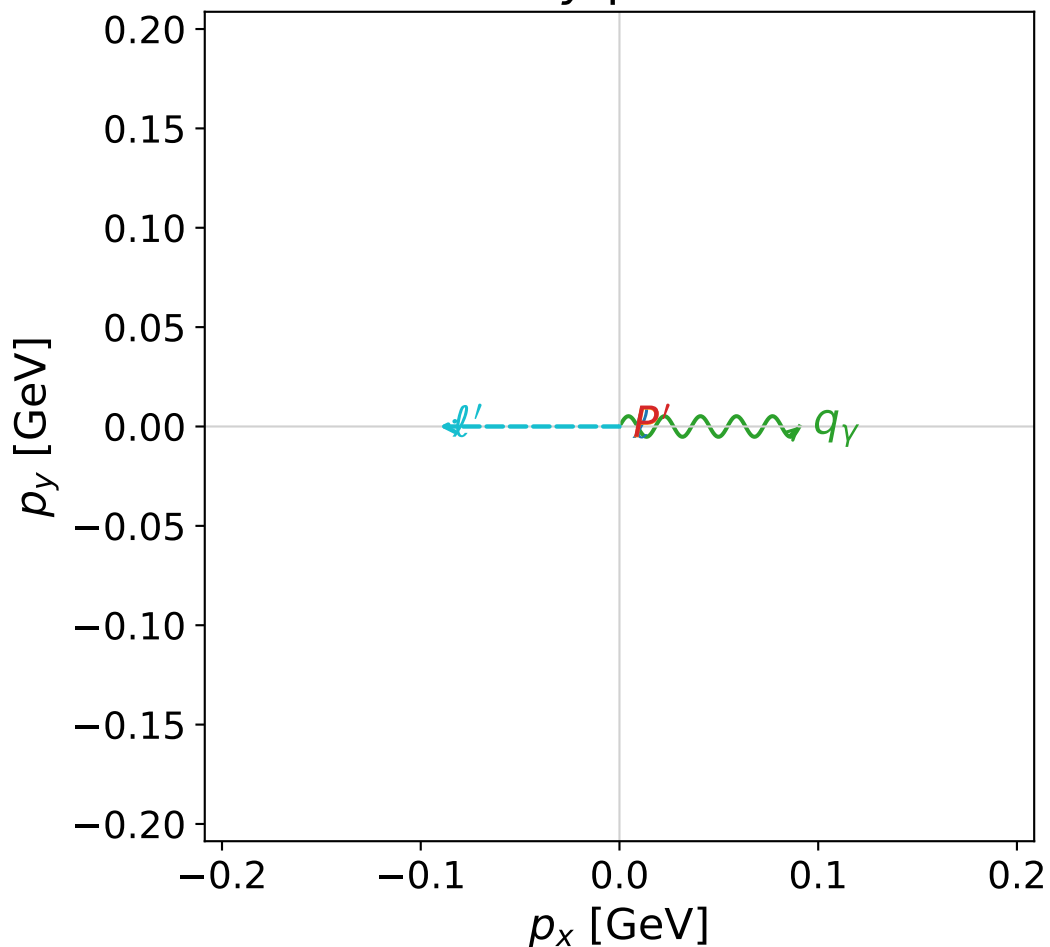
P $E= 0.9540$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1739$

ℓ' $E= 0.0958$ $p_x= -0.0906$ $p_y= -0.0000$ $p_z= -0.0312$

P' $E= 0.9380$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.0059$

q_γ $E= 0.0941$ $p_x= 0.0906$ $p_y= 0.0000$ $p_z= 0.0253$

x--y plane



Leading normalized final-state amplitudes
(N=3, retained=97.5%)

